

MBIE Housing Affordability Project — Regression Analysis

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Contents

1	Analysis Overview	5
2	1. Canterbury Housing Affordability	5
2.1	Why this dataset is used	5
2.2	Preparing the Canterbury affordability data	7
2.3	Data quality checks	8
2.4	Finding	9
3	2. Descriptive Statistics for Canterbury Affordability	9
3.1	Why this analysis is used	9
3.2	Interpretation	10
4	3. House Value-to-Income Over Time	10
4.1	Why this analysis is used	10
4.2	Interpretation	11
5	4. Mortgage Payment Burden Over Time	11
5.1	Why this analysis is used	11
5.2	Interpretation	12
6	5. Largest Changes in House Value-to-Income	12
6.1	Interpretation	13
7	6. Largest Changes in Mortgage Burden	13
7.1	Interpretation	14
8	7. Relationship Between the Two Affordability Measures	14
8.1	Interpretation	14

9	8. Recent Canterbury Affordability	15
9.1	Interpretation	15
10	9. Simple Time Trend — House Value-to-Income	16
10.1	Interpretation	16
11	10. Simple Time Trend — Mortgage Burden	16
11.1	Interpretation	17
12	11. Canterbury Dwelling Consents	17
12.1	Why dwelling consents were investigated	17
12.2	Restricting the data to 2021–2025	18
13	12. Exploratory Analysis of Dwelling Consents	19
13.1	Interpretation	21
14	13. Annual Dwelling Consent Trends	21
14.1	Interpretation	22
15	14. Dwelling Consents and Canterbury Affordability	22
15.1	Interpretation	24
16	15. Dwelling Consents and Mortgage Burden	24
16.1	Interpretation	26
17	16. Value of Consented Building Work per Square Metre	26
17.1	Why this variable is investigated	26
17.2	Interpretation	29
18	17. CPI and Inflation	29
18.1	Why inflation is investigated	29
18.2	Interpretation	31
19	18. Net Migration	31
19.1	Why migration is investigated	31
19.2	Interpretation	32
20	19. Mortgage Interest Rates	32
20.1	Why mortgage rates were investigated	32
20.2	Interpretation	33

21	20. Other Datasets Investigated	33
21.1	Population	34
21.2	QV data	34
21.3	Tenancy data	34
21.4	ECAN property data	34
21.5	Local Housing Statistics	35
22	21. Annual Canterbury House-Value Data	35
22.1	Interpretation	38
23	22. Quarterly HPI Outcome	39
24	23. Regression-Ready Master Dataset	40
24.1	Data-quality finding	45
25	24. Descriptive Statistics for the Quarterly Dataset	45
25.1	Interpretation	46
26	25. Correlation Analysis Using Levels	46
26.1	Individual correlation tests	47
27	26. HPI Trend	48
27.1	Interpretation	49
28	27. Scatterplots of HPI and Candidate Predictors	49
29	28. Creating Quarterly Changes and Lagged Variables	52
30	29. Quarterly HPI Growth	54
30.1	Interpretation	55
31	30. Correlations Using Quarterly Changes	55
31.1	Interpretation	56
32	31. Simple Regression Models	56
32.1	31.1 Migration Model	56
32.2	Interpretation	57
32.3	31.2 Housing Supply Model	57
32.4	Interpretation	58
32.5	31.3 Construction-Value Growth Model	58
32.6	Interpretation	59

33	32. Small Combined Regression	59
	33.1 Interpretation	60
34	33. HPI Persistence Regression	60
	34.1 Why this model is important	60
	34.2 Main Results	61
	34.3 Interpretation	61
35	34. Mixed Regression — Persistence and Migration	61
	35.1 Interpretation	62
36	35. Demand and Supply Regression	62
	36.1 Interpretation	63
37	36. Full Multivariable Regression	63
	37.1 Reason for model	63
	37.2 Main Results	64
	37.3 Interpretation	64
38	37. Model Comparison	65
39	38. Model Sample Sizes	65
40	39. Regression Diagnostics	66
	40.1 Interpretation	66
41	40. Residual Autocorrelation	67
	41.1 Interpretation	68
42	41. Main Findings from the Regression Analysis	68
43	42. How These Results Help the Project	69
44	43. What the Analysis Does Not Show	69
45	44. Limitations	69
46	45. Overall Conclusion	70

1 Analysis Overview

The purpose of my analysis is to investigate factors that may be associated with housing affordability and housing-market conditions, with particular attention to Canterbury and the recent 2021–2025 period.

I do not begin immediately with regression. I first investigate each potential explanatory variable to determine whether the dataset is suitable, whether its geographic and temporal coverage matches my research question, and whether there is enough information to justify including it in a statistical model.

The main Canterbury affordability measures I use are the **house value-to-income multiple** and **mortgage payment proportion of income**.

In my later quarterly regression analysis, I use the available **House Price Index (HPI)** as my housing-market outcome and investigate whether quarterly HPI growth is associated with previous HPI growth, migration, dwelling consenting activity and the value of consented construction work.

I interpret the results as statistical associations rather than causal effects.

2 1. Canterbury Housing Affordability

2.1 Why this dataset is used

Before investigating possible explanatory factors, I first want to understand how housing affordability itself has changed.

I use an affordability dataset that contains two useful Canterbury indicators:

1. **House value-to-income multiple**, which compares housing values with household income.
2. **Mortgage payment proportion of income**, which measures the share of income required for mortgage payments.

I treat these as capturing different dimensions of affordability. House values relative to income represent the purchase-price dimension, while mortgage burden also reflects financing conditions.

```
affordability <- read.csv(
  "house-value-to-income-multiple-and-mortgage-payment-proportion-of-income (1).csv",
  stringsAsFactors = FALSE,
  check.names = FALSE,
  fileEncoding = "UTF-8-BOM"
)

# Remove stray spaces at the ends of the column names in the source file
names(affordability) <- trimws(names(affordability))

tbl(data.frame(Column = names(affordability)))
```

Column

Year

Canterbury Region House value to income

Canterbury Region Mortgage payment proportion of income

New Zealand House value to income

New Zealand Mortgage payment proportion of income

Column
Christchurch City House value to income
Christchurch City Mortgage payment proportion of income
Selwyn District House value to income
Selwyn District Mortgage payment proportion of income

```
tbl(data.frame(Rows = nrow(affordability), Columns = ncol(affordability)))
```

Rows	Columns
22	9

```
tbl(struct_tbl(affordability))
```

Column	Type
Year	integer
Canterbury Region House value to income	numeric
Canterbury Region Mortgage payment proportion of income	numeric
New Zealand House value to income	numeric
New Zealand Mortgage payment proportion of income	numeric
Christchurch City House value to income	numeric
Christchurch City Mortgage payment proportion of income	numeric
Selwyn District House value to income	numeric
Selwyn District Mortgage payment proportion of income	numeric

Column	First_values
Year	2005, 2006, 2007
Canterbury Region House value to income	4.4, 4.8, 5.1
Canterbury Region Mortgage payment proportion of income	31.8, 35, 40
New Zealand House value to income	4.8, 5.2, 5.4
New Zealand Mortgage payment proportion of income	34.8, 37.7, 42.6
Christchurch City House value to income	4.5, 4.8, 5.1
Christchurch City Mortgage payment proportion of income	32.7, 35.3, 40.3
Selwyn District House value to income	4.5, 5.3, 5.8
Selwyn District Mortgage payment proportion of income	32.7, 38.5, 45.9

```
affordability_summary <- summary_tbl(affordability)
affordability_summary$Variable <- gsub(
  "Mortgage payment proportion of income", "Mortgage burden",
  gsub("House value to income", "Value-to-income", affordability_summary$Variable)
)
tbl(affordability_summary)
```

Variable	Min	Q1	Median	Mean
Year	2005.0	2010.250	2015.50	2015.500
Canterbury Region Value-to-income	4.2	4.700	4.80	4.986
Canterbury Region Mortgage burden	24.7	27.400	29.65	31.368
New Zealand Value-to-income	4.6	5.025	5.60	5.745
New Zealand Mortgage burden	28.0	32.925	34.90	35.891
Christchurch City Value-to-income	4.3	4.800	4.90	5.091
Christchurch City Mortgage burden	25.1	27.925	30.75	32.114
Selwyn District Value-to-income	4.3	4.700	4.80	4.986
Selwyn District Mortgage burden	23.5	27.000	29.80	31.645

Variable	Q3	Max
Year	2020.750	2026.0
Canterbury Region Value-to-income	5.400	6.5
Canterbury Region Mortgage burden	34.750	40.5
New Zealand Value-to-income	6.375	8.0
New Zealand Mortgage burden	37.475	45.0
Christchurch City Value-to-income	5.550	6.7
Christchurch City Mortgage burden	35.300	41.2
Selwyn District Value-to-income	5.275	6.6
Selwyn District Mortgage burden	36.000	45.9

2.2 Preparing the Canterbury affordability data

In this section I assume the source file contains the Canterbury affordability observations I used in my original analysis.

If the source file already contains the cleaned variables `Year`, `ValueIncome` and `MortgageBurden`, my code uses them directly.

```
if (all(c("Year", "ValueIncome", "MortgageBurden") %in% names(affordability))) {

  canterbury_affordability <- affordability %>%
    select(Year, ValueIncome, MortgageBurden) %>%
    arrange(Year)

} else if (all(c(
  "Year",
  "Canterbury Region House value to income",
  "Canterbury Region Mortgage payment proportion of income"
) %in% names(affordability))) {

  # The raw source file stores the Canterbury Region measures under long
  # column names, so they are renamed to the cleaned names used below.
  canterbury_affordability <- affordability %>%
    select(
      Year,
      ValueIncome = `Canterbury Region House value to income`,
      MortgageBurden = `Canterbury Region Mortgage payment proportion of income`
    )
}
```

```

    ) %>%
    arrange(Year)
} else {

  stop(
    paste(
      "The affordability CSV loaded successfully, but the cleaned columns",
      "Year, ValueIncome and MortgageBurden are not present.",
      "Use the same cleaned affordability CSV that was used for the original analysis."
    )
  )
}

canterbury_affordability <- canterbury_affordability %>%
  mutate(
    ValueIncomeChange = ValueIncome - lag(ValueIncome),
    ValueIncomePctChange =
      (ValueIncome / lag(ValueIncome) - 1) * 100,

    MortgageBurdenChange =
      MortgageBurden - lag(MortgageBurden),

    MortgageBurdenPctChange =
      (MortgageBurden / lag(MortgageBurden) - 1) * 100
  )

```

2.3 Data quality checks

```

tbl(data.frame(
  Rows = nrow(canterbury_affordability),
  Columns = ncol(canterbury_affordability)
))

```

Rows	Columns
22	7

```

tbl(data.frame(
  Column = names(canterbury_affordability),
  Missing_values = colSums(is.na(canterbury_affordability)),
  row.names = NULL
))

```

Column	Missing_values
Year	0
ValueIncome	0
MortgageBurden	0

Column	Missing_values
ValueIncomeChange	1
ValueIncomePctChange	1
MortgageBurdenChange	1
MortgageBurdenPctChange	1

```
tbl(data.frame(
  Duplicate_years = sum(duplicated(canterbury_affordability$Year)),
  First_year = min(canterbury_affordability$Year),
  Last_year = max(canterbury_affordability$Year)
))
```

Duplicate_years	First_year	Last_year
0	2005	2026

2.4 Finding

I found that the Canterbury affordability data I used in my original analysis contained complete annual observations from **2005 to 2026**, with no duplicate years for the selected measures.

This gives me a useful long-run view of Canterbury affordability.

3 2. Descriptive Statistics for Canterbury Affordability

3.1 Why this analysis is used

I use descriptive statistics to build a baseline understanding of the typical level and variation in Canterbury housing affordability.

```
canterbury_affordability %>%
  summarise(
    Mean_ValueIncome = mean(ValueIncome, na.rm = TRUE),
    Median_ValueIncome = median(ValueIncome, na.rm = TRUE),
    SD_ValueIncome = sd(ValueIncome, na.rm = TRUE),
    Min_ValueIncome = min(ValueIncome, na.rm = TRUE),
    Max_ValueIncome = max(ValueIncome, na.rm = TRUE),

    Mean_MortgageBurden = mean(MortgageBurden, na.rm = TRUE),
    Median_MortgageBurden = median(MortgageBurden, na.rm = TRUE),
    SD_MortgageBurden = sd(MortgageBurden, na.rm = TRUE),
    Min_MortgageBurden = min(MortgageBurden, na.rm = TRUE),
    Max_MortgageBurden = max(MortgageBurden, na.rm = TRUE)
  ) %>%
  pivot_longer(
    cols = everything(),
    names_to = "Measure",
```

```

  values_to = "Value"
) %>%
tbl()

```

Measure	Value
Mean_ValueIncome	4.986
Median_ValueIncome	4.800
SD_ValueIncome	0.577
Min_ValueIncome	4.200
Max_ValueIncome	6.500
Mean_MortgageBurden	31.368
Median_MortgageBurden	29.650
SD_MortgageBurden	5.223
Min_MortgageBurden	24.700
Max_MortgageBurden	40.500

3.2 Interpretation

I found that across 2005–2026, Canterbury’s house value-to-income multiple averaged approximately **4.99**, meaning that house values were approximately five times household income on average.

I also found that mortgage payments averaged approximately **31.37% of income**.

Both measures varied substantially over the period, which suggests to me that Canterbury housing affordability changed considerably rather than remaining stable.

Potential project finding: I found that between 2005 and 2026, Canterbury’s house value-to-income multiple averaged approximately 4.99, while mortgage payments averaged approximately 31.37% of income. Both measures varied substantially over time.

4 3. House Value-to-Income Over Time

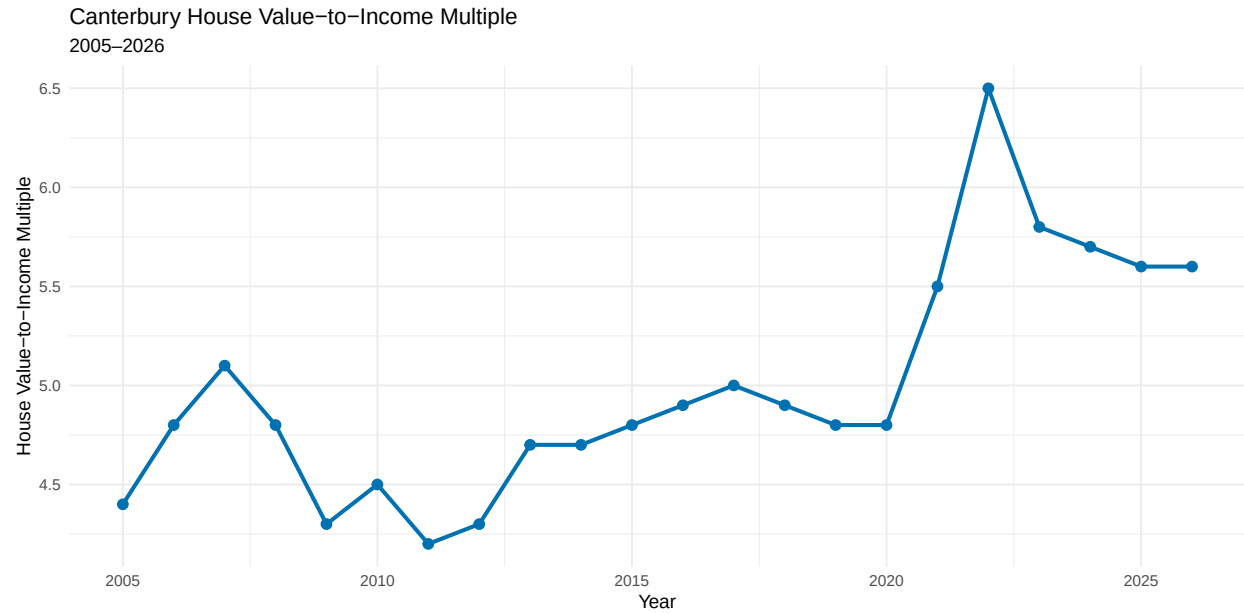
4.1 Why this analysis is used

The descriptive statistics show me variation but do not reveal **when** affordability changed.

```

ggplot(
  canterbury_affordability,
  aes(x = Year, y = ValueIncome)
) +
  geom_line(color = "#0072B2", linewidth = 1.1) +
  geom_point(color = "#0072B2", size = 2.5) +
  labs(
    title = "Canterbury House Value-to-Income Multiple",
    subtitle = "2005-2026",
    x = "Year",
    y = "House Value-to-Income Multiple"
  ) +
  theme_minimal()

```



4.2 Interpretation

From the graph, I can see approximately three broad periods.

2005–2020: I see that the ratio fluctuated but remained relatively stable compared with the later increase.

2021–2022: I see affordability deteriorating rapidly, with the ratio increasing from **4.8 in 2020 to 5.5 in 2021 and 6.5 in 2022**.

2023 onward: I see the ratio declining from its 2022 peak, indicating partial improvement, although it remained above the 2020 level.

Potential project finding: I found that Canterbury’s house value-to-income multiple increased sharply during 2021–2022, reaching a series maximum of 6.5 in 2022. Although the multiple subsequently declined, it remained above its 2020 level.

This makes the period around **2021–2023** particularly important for my later analysis.

5 4. Mortgage Payment Burden Over Time

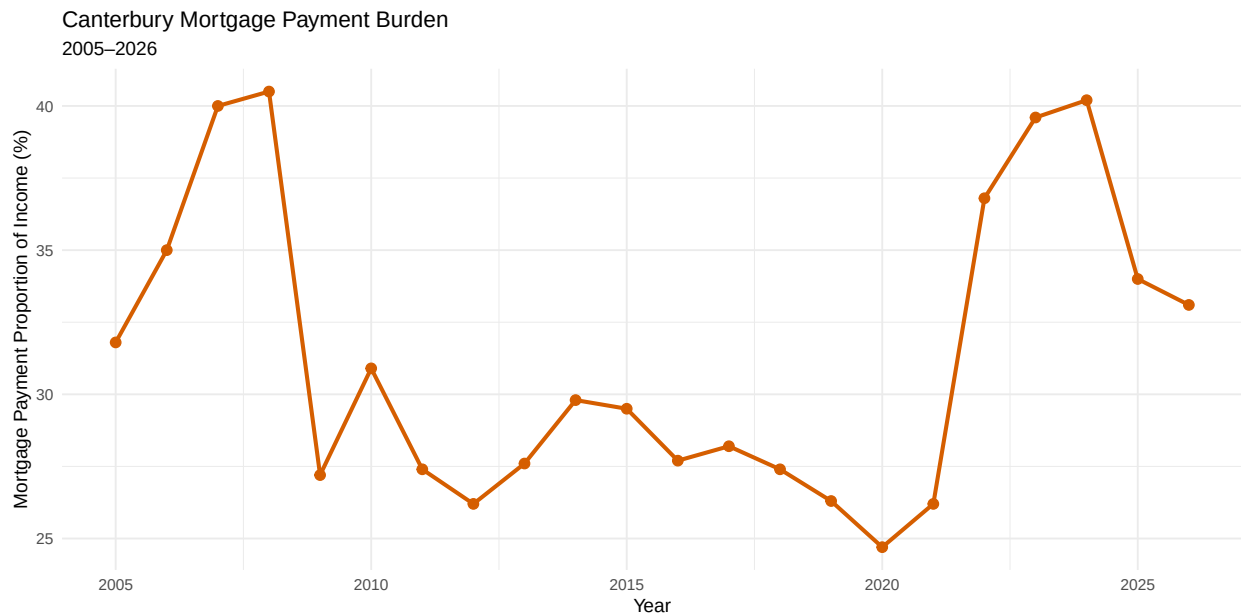
5.1 Why this analysis is used

I recognise that house value-to-income is only one dimension of affordability.

A household can experience increasing mortgage pressure even when house values stabilise or decline because mortgage payments also depend on financing conditions.

```
ggplot(
  canterbury_affordability,
  aes(x = Year, y = MortgageBurden)
) +
  geom_line(color = "#D55E00", linewidth = 1.1) +
```

```
geom_point(color = "#D55E00", size = 2.5) +
labs(
  title = "Canterbury Mortgage Payment Burden",
  subtitle = "2005-2026",
  x = "Year",
  y = "Mortgage Payment Proportion of Income (%)"
) +
theme_minimal()
```



5.2 Interpretation

I notice that mortgage burden behaves differently from the house value-to-income multiple.

I see that it was relatively high during 2007–2008, declined considerably afterward and reached **24.7% in 2020**.

I then see it increase sharply, reaching **40.2% in 2024**, before declining to **34.0% in 2025** and **33.1% in 2026**.

Potential project finding: I found that mortgage payment burden in Canterbury did not follow a simple long-term upward pattern. Instead, the series experienced distinct periods of high and low burden, including relatively high levels around 2007–2008 and again during 2022–2024.

This gives me a reason to investigate changing economic and financing conditions rather than assuming that affordability is explained simply by time.

6 5. Largest Changes in House Value-to-Income

```
canterbury_affordability %>%
  filter(!is.na(ValueIncomePctChange)) %>%
  arrange(desc(abs(ValueIncomePctChange))) %>%
  select(
    Year,
    ValueIncome,
    ValueIncomePctChange
  ) %>%
  slice_head(n = 6) %>%
  tbl()
```

Year	ValueIncome	ValueIncomePctChange
2022	6.5	18.182
2021	5.5	14.583
2023	5.8	-10.769
2009	4.3	-10.417
2013	4.7	9.302
2006	4.8	9.091

6.1 Interpretation

I find that the two largest annual increases occur consecutively in **2021 and 2022**, followed by a substantial decrease in 2023.

Potential project finding: I found that the most pronounced deterioration in Canterbury's house value-to-income measure occurred during 2021–2022, with annual increases of approximately 14.6% and 18.2%, respectively. This was followed by an approximately 10.8% decrease in 2023.

This leads me to ask what changed around 2021–2023.

7 6. Largest Changes in Mortgage Burden

```
canterbury_affordability %>%
  filter(!is.na(MortgageBurdenPctChange)) %>%
  arrange(desc(abs(MortgageBurdenPctChange))) %>%
  select(
    Year,
    MortgageBurden,
    MortgageBurdenChange,
    MortgageBurdenPctChange
  ) %>%
  slice_head(n = 6) %>%
  tbl()
```

Year	MortgageBurden	MortgageBurdenChange	MortgageBurdenPctChange
2022	36.8	10.6	40.458
2009	27.2	-13.3	-32.840
2025	34.0	-6.2	-15.423
2007	40.0	5.0	14.286
2010	30.9	3.7	13.603
2011	27.4	-3.5	-11.327

7.1 Interpretation

I find the 2022 change particularly notable.

I see mortgage burden increase from **26.2% to 36.8%**, an increase of **10.6 percentage points in one year**.

At this stage, I do not want to automatically describe this as an interest-rate effect. Instead, it gives me a reason to investigate financing conditions.

8 7. Relationship Between the Two Affordability Measures

```
affordability_cor_test <- cor.test(
  canterbury_affordability$ValueIncome,
  canterbury_affordability$MortgageBurden,
  method = "pearson"
)

tbl(
  cor_row(affordability_cor_test, "ValueIncome vs MortgageBurden"),
  digits = 4
)
```

Test	r	t	df	p_value	CI_Lower
ValueIncome vs MortgageBurden	0.538	2.8541	20	0.0098	0.1505

Test	CI_Upper
ValueIncome vs MortgageBurden	0.7822

8.1 Interpretation

My original analysis produced a Pearson correlation of approximately:

$$r = 0.538$$

with:

$$p = 0.0098$$

I interpret this as a moderate positive sample association: years with higher house value-to-income multiples generally also had higher mortgage payment burdens.

However, my observations form a time series, so I recognise that the standard independence assumption behind the Pearson test may not fully hold.

I therefore treat the result as exploratory rather than causal evidence.

9 8. Recent Canterbury Affordability

```
affordability_5yr <- canterbury_affordability %>%  
  filter(Year >= 2021, Year <= 2025) %>%  
  select(  
    Year,  
    ValueIncome,  
    MortgageBurden  
  )  
  
tbl(affordability_5yr)
```

Year	ValueIncome	MortgageBurden
2021	5.5	26.2
2022	6.5	36.8
2023	5.8	39.6
2024	5.7	40.2
2025	5.6	34.0

9.1 Interpretation

I notice an important divergence after 2022.

House value-to-income improved:

$$6.5 \rightarrow 5.8 \rightarrow 5.7$$

while mortgage burden increased:

$$36.8\% \rightarrow 39.6\% \rightarrow 40.2\%$$

Potential project finding: I found that the two affordability measures diverged after 2022. While Canterbury's house value-to-income multiple declined from 6.5 in 2022 to 5.7 in 2024, mortgage payment burden increased from 36.8% to 40.2%.

This suggests to me that purchase-price affordability and mortgage-payment affordability were being influenced by different conditions.

10 9. Simple Time Trend — House Value-to-Income

```
value_time_model <- lm(
  ValueIncome ~ Year,
  data = canterbury_affordability
)

coef_tbl(value_time_model)
```

Term	Estimate	Std_Err	t_value	p_value	CI_Low	CI_High
(Intercept)	-127.1	27.01	-4.707	0.0001351	-183.5	-70.8
Year	0.06556	0.0134	4.892	8.814e-05	0.0376	0.09351

```
fit_tbl(value_time_model)
```

N	R2	Adj_R2	Resid_SE	F_stat	p_value
22	0.5447	0.522	0.3988	23.93	8.814e-05

10.1 Interpretation

My original model estimated:

$$\beta_{Year} \approx 0.0656$$

with:

$$p < 0.001$$

and:

$$R^2 \approx 0.545$$

I therefore identify an overall upward linear trend in Canterbury's house value-to-income multiple over 2005–2026.

However, the graph also shows me substantial departures from a straight line, especially during the early 2020s.

I therefore treat time as a useful baseline that does not explain all of the observed movements.

11 10. Simple Time Trend — Mortgage Burden


```
mortgage_time_model <- lm(
  MortgageBurden ~ Year,
  data = canterbury_affordability
)

coef_tbl(mortgage_time_model)
```

Term	Estimate	Std_Err	t_value	p_value	CI_Low	CI_High
(Intercept)	-14.27	362.3	-0.03938	0.969	-770.1	741.5
Year	0.02264	0.1798	0.126	0.901	-0.3523	0.3976

```
fit_tbl(mortgage_time_model)
```

N	R2	Adj_R2	Resid_SE	F_stat	p_value
22	0.0007926	-0.04917	5.349	0.01586	0.901

11.1 Interpretation

My original analysis found:

$$\beta_{Year} \approx 0.0226$$

$$p = 0.901$$

$$R^2 \approx 0.001$$

I therefore find essentially no simple linear relationship between calendar year and mortgage burden over the full period.

This supports my decision to investigate changing economic and financing conditions rather than relying on time alone.

12 11. Canterbury Dwelling Consents

12.1 Why dwelling consents were investigated

I investigated dwelling consents because housing supply may affect the balance between housing demand and available housing.

Quarterly dwelling-consent data give me more observations than the annual affordability data, so I can examine 2021–2025 more closely.

My code below uses a normal relative file path rather than `file.choose()`.

```

consents <- readxl::read_excel(
  "Quarterly New Dwelling Consent Analysis Canterbury.xlsx"
)

names(consents)[1:7] <- c(
  "Quarter",
  "Dwellings",
  "AvgFloorArea",
  "ValuePerSQM",
  "TotalValue",
  "TotalFloorArea",
  "ValuePerConsent"
)

consents <- consents %>%
  mutate(
    Quarter = as.character(Quarter)
  )

tbl(struct_tbl(consents))

```

Column	Type	First_values
Quarter	character	2010Q1, 2010Q2, 2010Q3
Dwellings	numeric	756, 735, 653
AvgFloorArea	numeric	198, 205, 204
ValuePerSQM	numeric	1296, 1307, 1294
TotalValue	numeric	193849594, 197133986, 172577831
TotalFloorArea	numeric	149581, 150775, 133349
ValuePerConsent	numeric	256415, 268210, 264285

12.2 Restricting the data to 2021–2025

```

consents_5yr <- consents %>%
  filter(
    Quarter >= "2021Q1",
    Quarter <= "2025Q4"
  ) %>%
  mutate(
    Year = as.numeric(substr(Quarter, 1, 4))
  )

tbl(struct_tbl(consents_5yr))

```

Column	Type	First_values
Quarter	character	2021Q1, 2021Q2, 2021Q3
Dwellings	numeric	1573, 1985, 2149
AvgFloorArea	numeric	161, 156, 156

Column	Type	First_values
ValuePerSQM	numeric	2187, 2145, 2191
TotalValue	numeric	554957428, 663572561, 736893624
TotalFloorArea	numeric	253757, 309348, 336280
ValuePerConsent	numeric	352802, 334293, 342901
Year	numeric	2021, 2021, 2021

```
tbl(data.frame(Rows = nrow(consents_5yr)))
```

Rows
20

My resulting dataset contains **20 quarterly observations**, covering four quarters for each year from 2021 through 2025.

13 12. Exploratory Analysis of Dwelling Consents

```
consents_5yr %>%
  summarise(
    Mean = mean(Dwellings, na.rm = TRUE),
    Median = median(Dwellings, na.rm = TRUE),
    SD = sd(Dwellings, na.rm = TRUE),
    Minimum = min(Dwellings, na.rm = TRUE),
    Maximum = max(Dwellings, na.rm = TRUE)
  ) %>%
  tbl()
```

Mean	Median	SD	Minimum	Maximum
1871.55	1858	302.208	1305	2416

```
consents_5yr %>%
  filter(Dwellings == max(Dwellings, na.rm = TRUE)) %>%
  tbl()
```

Quarter	Dwellings	AvgFloorArea	ValuePerSQM	TotalValue
2022Q1	2416	148	2342	836466019

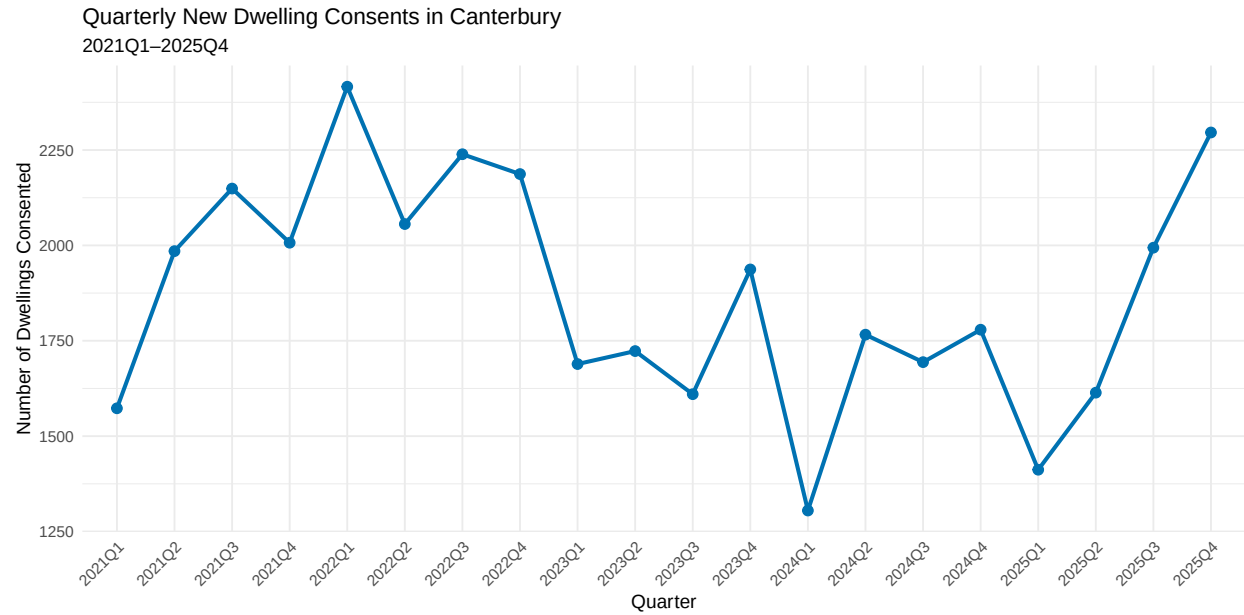
Quarter	TotalFloorArea	ValuePerConsent	Year
2022Q1	357167	346219	2022

```
consents_5yr %>%
  filter(Dwellings == min(Dwellings, na.rm = TRUE)) %>%
  tbl()
```

Quarter	Dwellings	AvgFloorArea	ValuePerSQM	TotalValue
2024Q1	1305	149	2867	556731415

Quarter	TotalFloorArea	ValuePerConsent	Year
2024Q1	194181	426614	2024

```
ggplot(
  consents_5yr,
  aes(x = Quarter, y = Dwellings, group = 1)
) +
  geom_line(
    color = "#0072B2",
    linewidth = 1.1
  ) +
  geom_point(
    color = "#0072B2",
    size = 2.5
  ) +
  labs(
    title = "Quarterly New Dwelling Consents in Canterbury",
    subtitle = "2021Q1-2025Q4",
    x = "Quarter",
    y = "Number of Dwellings Consented"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```



13.1 Interpretation

I find that quarterly consenting activity varies substantially.

The highest quarterly level I found was **2,416 dwellings in 2022Q1**, while the lowest was **1,305 dwellings in 2024Q1**.

This indicates to me that housing development activity was not constant during the study period.

14 13. Annual Dwelling Consent Trends

```
annual_consents <- consents_5yr %>%
  group_by(Year) %>%
  summarise(
    AnnualDwellings = sum(Dwellings, na.rm = TRUE),
    AvgQuarterlyDwellings = mean(Dwellings, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(Year) %>%
  mutate(
    AnnualChange = AnnualDwellings - lag(AnnualDwellings),
    PercentChange =
      (AnnualDwellings / lag(AnnualDwellings) - 1) * 100
  )
tbl(annual_consents)
```

Year	AnnualDwellings	AvgQuarterlyDwellings	AnnualChange
2021	7714	1928.50	NA

Year	AnnualDwellings	AvgQuarterlyDwellings	AnnualChange
2022	8898	2224.50	1184
2023	6959	1739.75	-1939
2024	6544	1636.00	-415
2025	7316	1829.00	772

Year	PercentChange
2021	NA
2022	15.349
2023	-21.791
2024	-5.964
2025	11.797

14.1 Interpretation

I found that Canterbury recorded **7,714 new dwelling consents in 2021**, increasing by approximately **15.4% to 8,898 in 2022**.

I then see consenting activity decline by approximately **21.8% in 2023** and another **6.0% in 2024**, before recovering by approximately **11.8% in 2025** to 7,316 dwellings.

I keep in mind that dwelling consents represent approvals for future construction rather than completed dwellings. I therefore expect that any relationship with housing conditions may occur with a lag.

15 14. Dwelling Consents and Canterbury Affordability

```
supply_affordability <- annual_consents %>%
  left_join(
    affordability_5yr,
    by = "Year"
  )

tbl(supply_affordability)
```

Year	AnnualDwellings	AvgQuarterlyDwellings	AnnualChange
2021	7714	1928.50	NA
2022	8898	2224.50	1184
2023	6959	1739.75	-1939
2024	6544	1636.00	-415
2025	7316	1829.00	772

Year	PercentChange	ValueIncome	MortgageBurden
2021	NA	5.5	26.2
2022	15.349	6.5	36.8
2023	-21.791	5.8	39.6
2024	-5.964	5.7	40.2
2025	11.797	5.6	34.0

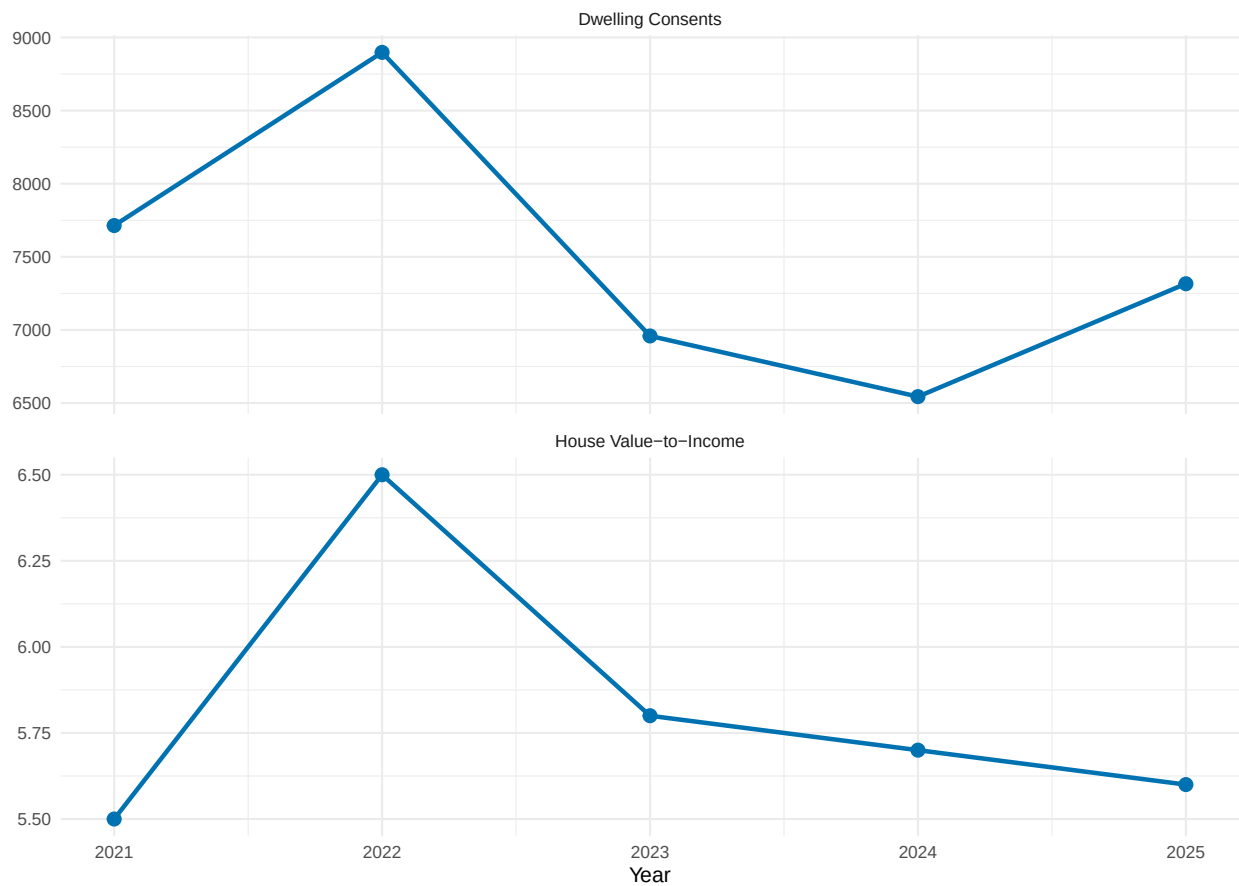
```

supply_plot_data <- supply_affordability %>%
  select(Year, AnnualDwellings, ValueIncome) %>%
  pivot_longer(
    cols = c(AnnualDwellings, ValueIncome),
    names_to = "Measure",
    values_to = "Value"
  ) %>%
  mutate(
    Measure = recode(
      Measure,
      AnnualDwellings = "Dwelling Consents",
      ValueIncome = "House Value-to-Income"
    )
  )

ggplot(
  supply_plot_data,
  aes(x = Year, y = Value)
) +
  geom_line(aes(group = Measure), color = "#0072B2", linewidth = 1.1) +
  geom_point(color = "#0072B2", size = 3) +
  facet_wrap(~ Measure, scales = "free_y", ncol = 1) +
  scale_x_continuous(breaks = 2021:2025) +
  labs(
    title = "Canterbury Dwelling Consents and Housing Affordability",
    subtitle = "2021-2025",
    x = "Year",
    y = NULL
  ) +
  theme_minimal()

```

Canterbury Dwelling Consents and Housing Affordability 2021–2025



15.1 Interpretation

I do not see a simple immediate same-year relationship in this comparison.

I see that both consenting and the value-to-income multiple increased from 2021 to 2022. From 2022 to 2024, both then declined.

I do not interpret this as evidence that supply is unimportant, because consenting represents planned future construction and may influence the market with a delay.

16 15. Dwelling Consents and Mortgage Burden

```
mortgage_supply_plot <- supply_affordability %>%  
  select(  
    Year,  
    AnnualDwellings,  
    MortgageBurden  
  ) %>%  
  pivot_longer(  
    cols = c(AnnualDwellings, MortgageBurden),
```



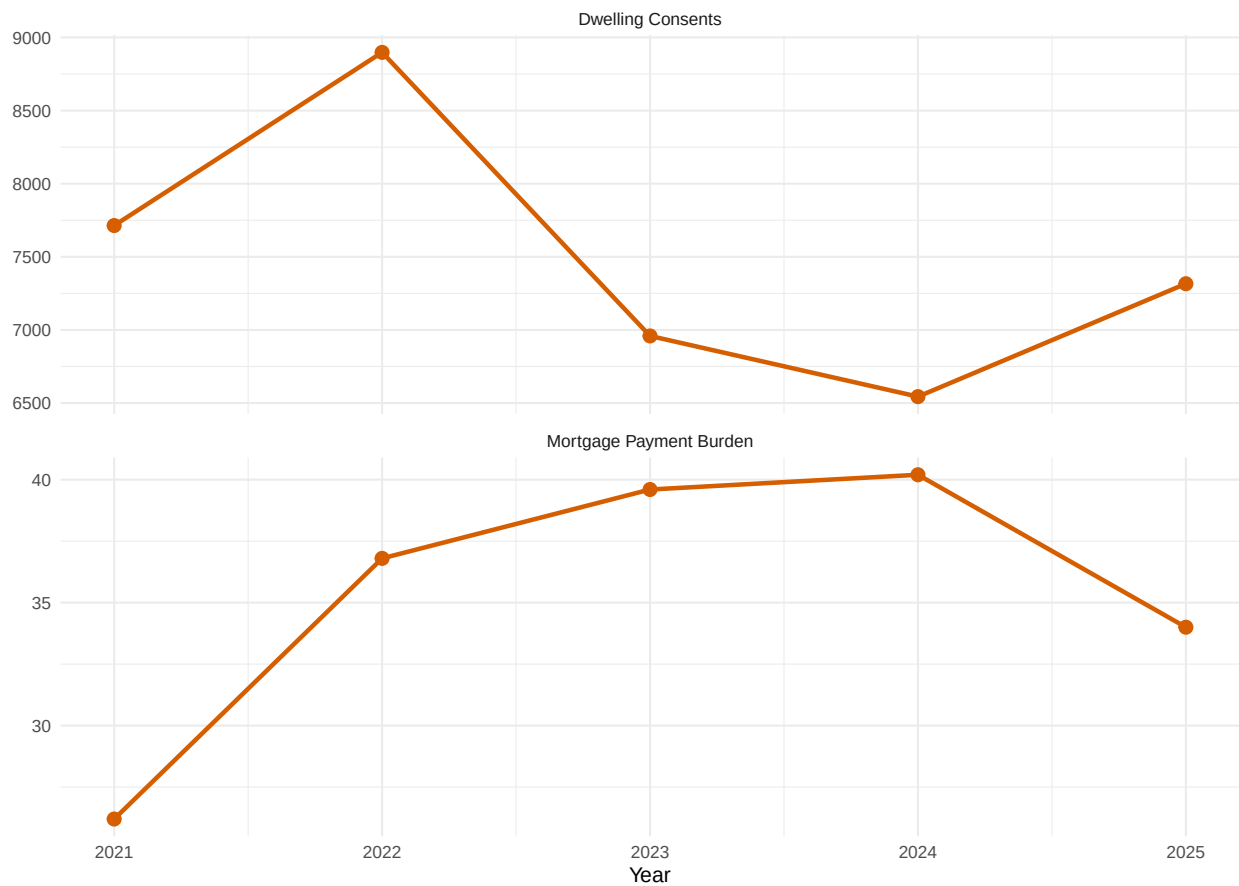
```

    names_to = "Measure",
    values_to = "Value"
  ) %>%
  mutate(
    Measure = recode(
      Measure,
      AnnualDwellings = "Dwelling Consents",
      MortgageBurden = "Mortgage Payment Burden"
    )
  )

ggplot(
  mortgage_supply_plot,
  aes(x = Year, y = Value)
) +
  geom_line(aes(group = Measure), color = "#D55E00", linewidth = 1.1) +
  geom_point(color = "#D55E00", size = 3) +
  facet_wrap(~ Measure, scales = "free_y", ncol = 1) +
  scale_x_continuous(breaks = 2021:2025) +
  labs(
    title = "Canterbury Dwelling Consents and Mortgage Burden",
    subtitle = "2021-2025",
    x = "Year",
    y = NULL
  ) +
  theme_minimal()

```

Canterbury Dwelling Consents and Mortgage Burden 2021–2025



16.1 Interpretation

Again, I do not see a consistent same-year relationship in the descriptive evidence.

I see that consents increased in 2022 while mortgage burden also increased sharply. Consenting then declined through 2024 while mortgage burden continued rising.

This suggests to me that dwelling consenting alone is not sufficient to describe recent mortgage-affordability changes.

17 16. Value of Consented Building Work per Square Metre

17.1 Why this variable is investigated

I use `ValuePerSQM` to represent the average value of consented residential building work per square metre.

I use it because it captures a different aspect of development activity from the number of dwellings alone.

I do not interpret it as a complete construction-cost index.

```
consents_5yr %>%
  summarise(
    Mean_ValuePerSQM = mean(ValuePerSQM, na.rm = TRUE),
    Median_ValuePerSQM = median(ValuePerSQM, na.rm = TRUE),
    SD_ValuePerSQM = sd(ValuePerSQM, na.rm = TRUE),
    Minimum_ValuePerSQM = min(ValuePerSQM, na.rm = TRUE),
    Maximum_ValuePerSQM = max(ValuePerSQM, na.rm = TRUE)
  ) %>%
  tbl()
```

Mean_ValuePerSQM	Median_ValuePerSQM	SD_ValuePerSQM
2668.8	2831	300.087

Mean_ValuePerSQM	Minimum_ValuePerSQM	Maximum_ValuePerSQM
2668.8	2145	2991

```
consents_5yr %>%
  filter(ValuePerSQM == min(ValuePerSQM, na.rm = TRUE)) %>%
  select(Quarter, ValuePerSQM) %>%
  tbl()
```

Quarter	ValuePerSQM
2021Q2	2145

```
consents_5yr %>%
  filter(ValuePerSQM == max(ValuePerSQM, na.rm = TRUE)) %>%
  select(Quarter, ValuePerSQM) %>%
  tbl()
```

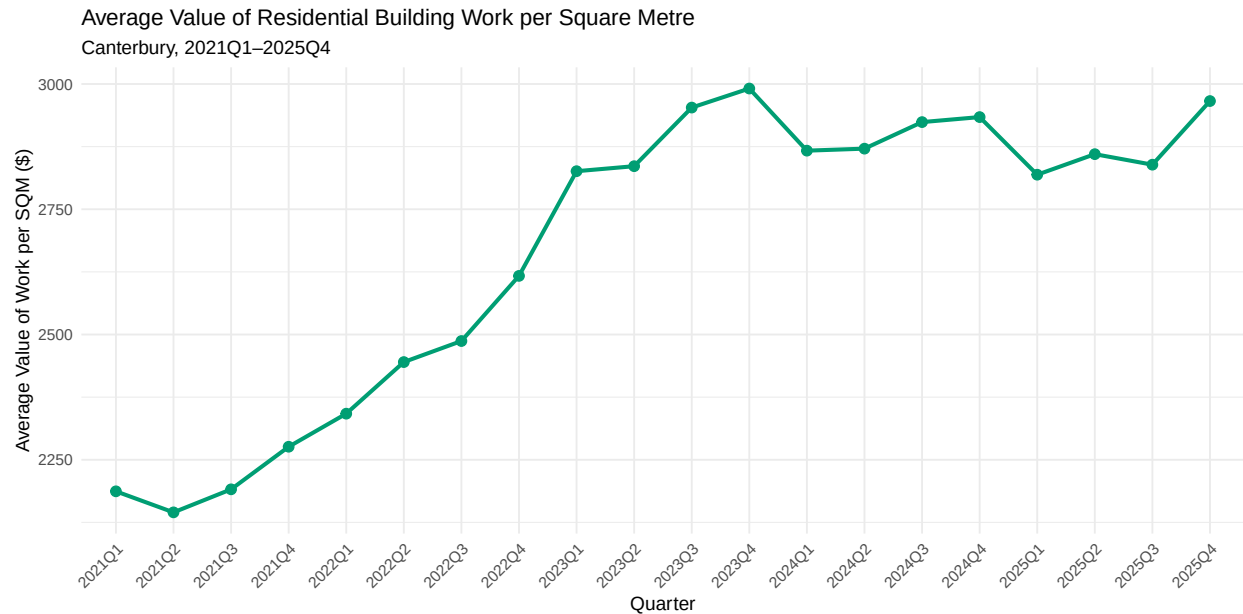
Quarter	ValuePerSQM
2023Q4	2991

```
ggplot(
  consents_5yr,
  aes(x = Quarter, y = ValuePerSQM, group = 1)
) +
  geom_line(color = "#009E73", linewidth = 1.1) +
  geom_point(color = "#009E73", size = 2.5) +
  labs(
    title = "Average Value of Residential Building Work per Square Metre",
    subtitle = "Canterbury, 2021Q1-2025Q4",
  )
```

```

x = "Quarter",
y = "Average Value of Work per SQM ($)"
) +
theme_minimal() +
theme(
  axis.text.x = element_text(angle = 45, hjust = 1)
)

```



```

annual_building_value <- consents_5yr %>%
  group_by(Year) %>%
  summarise(
    AvgValuePerSQM = mean(ValuePerSQM, na.rm = TRUE),
    AvgValuePerConsent = mean(ValuePerConsent, na.rm = TRUE),
    AvgFloorArea = mean(AvgFloorArea, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(Year) %>%
  mutate(
    ValuePerSQM_PctChange =
      (AvgValuePerSQM / lag(AvgValuePerSQM) - 1) * 100,

    ValuePerConsent_PctChange =
      (AvgValuePerConsent / lag(AvgValuePerConsent) - 1) * 100,

    FloorArea_PctChange =
      (AvgFloorArea / lag(AvgFloorArea) - 1) * 100
  )

tbl(annual_building_value)

```

Year	AvgValuePerSQM	AvgValuePerConsent	AvgFloorArea
2021	2199.75	344026.2	156.25
2022	2472.75	366394.8	148.25
2023	2901.50	407025.2	140.50
2024	2899.00	408090.2	141.00
2025	2871.00	405387.5	141.25

Year	ValuePerSQM_PctChange	ValuePerConsent_PctChange
2021	NA	NA
2022	12.411	6.502
2023	17.339	11.089
2024	-0.086	0.262
2025	-0.966	-0.662

Year	FloorArea_PctChange
2021	NA
2022	-5.120
2023	-5.228
2024	0.356
2025	0.177

17.2 Interpretation

I found that across 2021Q1–2025Q4, the average value of consented residential building work was approximately **\$2,669 per square metre**, with a standard deviation of approximately \$300.

The lowest quarterly value I found was approximately **\$2,145 in 2021Q2**, while the highest was approximately **\$2,991 in 2023Q4**.

I see the measure increase strongly between 2021 and 2023 and remain relatively high afterward.

I also see average floor area generally decline during this period, which suggests that the increase in value per square metre cannot simply be explained by larger dwellings.

I recognise that composition effects may still be present.

18 17. CPI and Inflation

18.1 Why inflation is investigated

I investigate inflation because it changed substantially during the same period as the housing market.

In my original analysis I converted CPI into year-on-year inflation rather than using only the trending CPI level.

The CPI workbook is read from the same folder as this R Markdown file. No file-selection window is used.

```

# The CPI workbook is located by its name pattern in the project folder, so
# small differences in the exact filename (e.g. "_csv" or "(1)") do not matter.
cpi_file <- list.files(pattern = "^consumers-price-index.*\\.xlsx$")[1]

if (is.na(cpi_file)) {
  stop(
    "No CPI workbook found in ", getwd(),
    ". Put the 'consumers-price-index...xlsx' file in the same folder as this .Rmd."
  )
}

cpi <- readxl::read_excel(
  cpi_file,
  sheet = excel_sheets(cpi_file)[1]
)

cpi_reg <- cpi %>%
  filter(
    Series_title_1 == "All groups",
    Period >= 2020.03,
    Period <= 2025.12
  ) %>%
  mutate(
    Quarter = case_when(
      grepl("\\.03$", as.character(Period)) ~ paste0(floor(Period), "Q1"),
      grepl("\\.06$", as.character(Period)) ~ paste0(floor(Period), "Q2"),
      grepl("\\.09$", as.character(Period)) ~ paste0(floor(Period), "Q3"),
      grepl("\\.12$", as.character(Period)) ~ paste0(floor(Period), "Q4")
    ),
    CPI = as.numeric(Data_value),
    Inflation_YoY = (CPI / lag(CPI, 4) - 1) * 100
  ) %>%
  filter(Period >= 2021.03) %>%
  select(Quarter, CPI, Inflation_YoY)

tbl(cpi_reg)

```

Quarter	CPI	Inflation_YoY
2021Q1	1068	1.521
2021Q2	1082	3.343
2021Q3	1106	4.934
2021Q4	1122	5.949
2022Q1	1142	6.929
2022Q2	1161	7.301
2022Q3	1186	7.233
2022Q4	1203	7.219
2023Q1	1218	6.655
2023Q2	1231	6.029
2023Q3	1253	5.649
2023Q4	1259	4.655
2024Q1	1267	4.023
2024Q2	1272	3.331

Quarter	CPI	Inflation_YoY
2024Q3	1280	2.155
2024Q4	1287	2.224
2025Q1	1299	2.526
2025Q2	1306	2.673
2025Q3	1319	3.047
2025Q4	1327	3.108

18.2 Interpretation

My original CPI analysis found that inflation increased substantially during 2021 and exceeded **7% during 2022**.

I then saw it decline through 2023 and 2024 before increasing moderately during 2025.

Because the CPI series is national, I interpret it as a national macroeconomic condition rather than a Canterbury-specific inflation measure.

19 18. Net Migration

19.1 Why migration is investigated

I investigate migration because it may affect housing demand by changing the number of people requiring accommodation.

In my original analysis I used a New Zealand net-migration measure.

```
migration <- readxl::read_excel(
  "Migration_NZ.xlsx",
  sheet = "Sheet1"
)

migration_q <- migration %>%
  mutate(
    Year = year(`Year ended`),
    Month = month(`Year ended`),
    Quarter = paste0(
      Year,
      "Q",
      quarter(`Year ended`)
    )
  ) %>%
  filter(
    Year >= 2021,
    Year <= 2025,
    Month %in% c(3, 6, 9, 12)
  ) %>%
  select(
    Quarter,
    NetMigration_NZ = `Net migration`
```

```
)  
tbl(migration_q)
```

Quarter	NetMigration_NZ
2021Q1	-1782
2021Q2	-6585
2021Q3	-10821
2021Q4	-14950
2022Q1	-19458
2022Q2	-17692
2022Q3	-5169
2022Q4	24900
2023Q1	74466
2023Q2	108397
2023Q3	132723
2023Q4	127951
2024Q1	100024
2024Q2	70216
2024Q3	42233
2024Q4	23880
2025Q1	14048
2025Q2	11319
2025Q3	12211
2025Q4	18462

19.2 Interpretation

My original migration series showed negative net migration during parts of 2021–2022 and much stronger positive migration during 2023.

The variable represents **New Zealand net migration rather than Canterbury-specific migration**, so I must consider this geographic limitation.

20 19. Mortgage Interest Rates

20.1 Why mortgage rates were investigated

Canterbury mortgage burden increased substantially after 2021 and continued rising even after the house value-to-income multiple began falling.

This gives me a strong reason to investigate financing conditions.

```
mortgage <- read.csv(  
  "New residential mortgage weighted average interest rates (selected terms) - B30.csv",  
  stringsAsFactors = FALSE,  
  fileEncoding = "UTF-8-BOM"  
)
```



```

mortgage_q <- mortgage %>%
  mutate(
    Date = dmy(Date),
    Year = year(Date),
    Quarter = paste0(Year, "Q", quarter(Date))
  ) %>%
  filter(
    Year >= 2021,
    Year <= 2025
  ) %>%
  group_by(Quarter) %>%
  summarise(
    MortgageRate_1yr = mean(X1.year.rate, na.rm = TRUE),
    .groups = "drop"
  )

tbl(mortgage_q)

```

Quarter	MortgageRate_1yr
2021Q2	2.297
2021Q3	2.407
2021Q4	2.940
2022Q1	3.457
2022Q2	3.997
2022Q3	4.770
2022Q4	5.340
2023Q1	6.057
2023Q2	6.330
2023Q3	6.737
2023Q4	7.007
2024Q1	7.010
2024Q2	6.863
2024Q3	6.617
2024Q4	5.793
2025Q1	5.393
2025Q2	5.003
2025Q3	4.780
2025Q4	4.423

20.2 Interpretation

My original analysis found that mortgage rates increased strongly from 2021 through 2023, remained high during 2024 and declined during 2025.

This timing is relevant to my earlier finding that mortgage burden remained high even after the value-to-income multiple started improving.

21 20. Other Datasets Investigated

I investigated several other datasets before choosing my final modelling approach.

21.1 Population

I investigated Canterbury population as a potential demand variable.

The available recent observations I had were approximately:

- 2023: 676,600
- 2024: 690,600
- 2025: 698,200

Only three overlapping recent observations were available, so I did not consider a separate regression based on population statistically reliable.

I therefore did not force population into the main regression.

21.2 QV data

I investigated a QV dataset as a possible higher-frequency housing outcome.

The available file did not provide sufficiently clear Canterbury-specific geography and date information for my intended regression and appeared to represent New Zealand overall.

I therefore did not substitute it for a Canterbury-specific outcome.

21.3 Tenancy data

I investigated a detailed tenancy dataset as a possible source of rental information.

The file contained location identifiers, but I did not identify a reliable lookup between those identifiers and Canterbury locations in the available project material.

I therefore did not use the dataset for geographic conclusions at this stage.

21.4 ECAN property data

I found that the ECAN property dataset contained approximately **328,883 records representing 328,763 unique valuation numbers**.

I found only around 120 records were duplicates by valuation number.

I therefore treated the dataset primarily as a **cross-sectional property-level dataset** rather than a longitudinal quarterly panel.

My initial exploration showed substantial differences in median property valuations across Canterbury councils, but these figures included multiple property categories, so I do not automatically interpret them as residential affordability differences.

I consider this dataset more suitable for the geographic component of the project than for the quarterly time-series regression.

21.5 Local Housing Statistics

I found that the Local Housing Statistics dataset contained useful rental-affordability indicators such as households spending at least 30%, 40% or 50% of income on rent.

However, the relevant observations were available for selected years such as 2001, 2006, 2013 and 2018 rather than quarterly during 2021–2025.

I therefore retained it as supplementary contextual evidence rather than as the dependent variable for the quarterly regression.

22 21. Annual Canterbury House-Value Data

I investigated several additional house-value datasets.

My code below reads the files directly from the project folder (the same folder as this R Markdown file) instead of a personal C:/Users/... path.

```
median_house <- read.csv(
  "median-average-house-value.csv",
  stringsAsFactors = FALSE
)

mean_house_1 <- read.csv(
  "mean-average-house-value (1).csv",
  stringsAsFactors = FALSE
)

mean_house_2 <- read.csv(
  "mean-average-house-value (2).csv",
  stringsAsFactors = FALSE
)

mean_house_3 <- read.csv(
  "mean-average-house-value (3).csv",
  stringsAsFactors = FALSE
)

mean_house <- read.csv(
  "mean-average-house-value.csv",
  stringsAsFactors = FALSE
)

house_files <- list(
  "median-average-house-value.csv" = median_house,
  "mean-average-house-value (1).csv" = mean_house_1,
  "mean-average-house-value (2).csv" = mean_house_2,
  "mean-average-house-value (3).csv" = mean_house_3,
  "mean-average-house-value.csv" = mean_house
)

tbl(data.frame(
  File = names(house_files),
  Rows = sapply(house_files, nrow),
```

```
Columns = sapply(house_files, ncol),
row.names = NULL
))
```

File	Rows	Columns
median-average-house-value.csv	9	7
mean-average-house-value (1).csv	21	3
mean-average-house-value (2).csv	22	2
mean-average-house-value (3).csv	22	7
mean-average-house-value.csv	2	2

```
tbl(data.frame(
  File = rep(names(house_files), sapply(house_files, ncol)),
  Column = unname(unlist(lapply(house_files, names))),
  row.names = NULL
))
```

File	Column
median-average-house-value.csv	Year
median-average-house-value.csv	Canterbury.Region.Level
median-average-house-value.csv	Canterbury.Region... Change
median-average-house-value.csv	Canterbury.Region.Absolute.change
median-average-house-value.csv	New.Zealand.Level
median-average-house-value.csv	New.Zealand... Change
median-average-house-value.csv	New.Zealand.Absolute.change
mean-average-house-value (1).csv	Year
mean-average-house-value (1).csv	Canterbury.Region
mean-average-house-value (1).csv	New.Zealand
mean-average-house-value (2).csv	Year
mean-average-house-value (2).csv	Canterbury.Region
mean-average-house-value (3).csv	Year
mean-average-house-value (3).csv	Canterbury.Region.Level
mean-average-house-value (3).csv	Canterbury.Region... Change
mean-average-house-value (3).csv	Canterbury.Region.Absolute.change
mean-average-house-value (3).csv	New.Zealand.Level
mean-average-house-value (3).csv	New.Zealand... Change
mean-average-house-value (3).csv	New.Zealand.Absolute.change
mean-average-house-value.csv	Category
mean-average-house-value.csv	Series.1

```
tbl(tidy_names(head(median_house, 10)))
```

Year	Cant Level	Cant Change	Cant Absolute change	NZ Level
2018	474307	NA	NA	623144
2019	476246	0.4	1939	639243

Year	Cant Level	Cant Change	Cant Absolute change	NZ Level
2020	494658	3.9	18412	683944
2021	573384	15.9	78726	819327
2022	747544	30.4	174160	968082
2023	668441	-10.6	-79103	810523
2024	706148	5.6	37707	841832
2025	711271	0.7	5123	817757
2026	732600	3.0	21329	812776

Year	NZ Change	NZ Absolute change
2018	NA	NA
2019	2.6	16099
2020	7.0	44701
2021	19.8	135383
2022	18.2	148755
2023	-16.3	-157559
2024	3.9	31309
2025	-2.9	-24075
2026	-0.6	-4981

```
tbl(tidy_names(head(mean_house_1, 10)))
```

Year	Cant	NZ
2006	13.6	12.2
2007	12.8	11.7
2008	3.1	2.8
2009	-8.2	-9.1
2010	6.1	6.3
2011	-0.6	-1.5
2012	7.1	3.2
2013	9.9	7.7
2014	6.7	7.8
2015	3.4	9.1

```
tbl(tidy_names(head(mean_house_2, 10)))
```

Year	Cant
2005	251854
2006	286107
2007	322604
2008	332674
2009	305392
2010	324132
2011	322298

Year	Cant
2012	345206
2013	379437
2014	404923

```
tbl(tidy_names(head(mean_house_3, 10)))
```

Year	Cant Level	Cant Change	Cant Absolute change	NZ Level
2005	251854	NA	NA	302302
2006	286107	13.6	34253	339202
2007	322604	12.8	36497	378950
2008	332674	3.1	10070	389733
2009	305392	-8.2	-27282	354277
2010	324132	6.1	18740	376446
2011	322298	-0.6	-1834	370730
2012	345206	7.1	22908	382579
2013	379437	9.9	34231	412150
2014	404923	6.7	25486	444261

Year	NZ Change	NZ Absolute change
2005	NA	NA
2006	12.2	36900
2007	11.7	39748
2008	2.8	10783
2009	-9.1	-35456
2010	6.3	22169
2011	-1.5	-5716
2012	3.2	11849
2013	7.7	29571
2014	7.8	32111

```
tbl(tidy_names(head(mean_house, 10)))
```

Category	Series 1
Canterbury Region	743171
New Zealand	870908

22.1 Interpretation

Inspecting these datasets showed me that the Canterbury house-value series were annual rather than quarterly.

The available annual series nevertheless gave me useful descriptive evidence.

In one Canterbury house-value series, I found that values increased from approximately **\$573,384 in 2021 to \$747,544 in 2022**, declined to approximately **\$668,441 in 2023**, and subsequently increased to approximately **\$706,148 in 2024 and \$711,271 in 2025**.

I found that a second annual series followed a similar pattern.

I do **not** duplicate these annual values across quarters because that would create artificial pseudo-observations and exaggerate the effective sample size.

23 22. Quarterly HPI Outcome

A quarterly HPI series covering 2021Q1–2025Q4 later became available to me for the housing-market regression.

```
housing_q <- data.frame(  
  Quarter = paste0(  
    rep(2021:2025, each = 4),  
    "Q",  
    rep(1:4, times = 5)  
  ),  
  
  HPI = c(  
    3379, 3495, 3704, 3920,  
    3846, 3681, 3548, 3484,  
    3382, 3349, 3404, 3464,  
    3470, 3398, 3381, 3414,  
    3412, 3375, 3376, 3392  
  )  
)  
  
tbl(housing_q)
```

Quarter	HPI
2021Q1	3379
2021Q2	3495
2021Q3	3704
2021Q4	3920
2022Q1	3846
2022Q2	3681
2022Q3	3548
2022Q4	3484
2023Q1	3382
2023Q2	3349
2023Q3	3404
2023Q4	3464
2024Q1	3470
2024Q2	3398
2024Q3	3381
2024Q4	3414
2025Q1	3412

Quarter	HPI
2025Q2	3375
2025Q3	3376
2025Q4	3392

This gives me **20 quarterly HPI observations** from 2021Q1 to 2025Q4.

24 23. Regression-Ready Master Dataset

My original completed master dataset contained:

- HPI
- New Zealand net migration
- CPI
- year-on-year inflation
- Canterbury dwelling consents
- average floor area
- value per square metre
- total consent value
- total floor area
- value per consent
- Canterbury annual value-to-income multiple
- Canterbury annual mortgage burden

To make my regression section fully reproducible even when the original CPI and migration workbook filenames are not known, I reconstruct the verified quarterly values from my completed master dataset below.

```
master_q <- data.frame(
  Quarter = paste0(
    rep(2021:2025, each = 4),
    "Q",
    rep(1:4, times = 5)
  ),

  HPI = c(
    3379, 3495, 3704, 3920,
    3846, 3681, 3548, 3484,
    3382, 3349, 3404, 3464,
    3470, 3398, 3381, 3414,
    3412, 3375, 3376, 3392
  ),

  NetMigration_NZ = c(
    -1782, -6585, -10821, -14950,
    -19458, -17692, -5169, 24900,
    74466, 108397, 132723, 127951,
    100024, 70216, 42233, 23880,
    14048, 11319, 12211, 18462
  ),
)
```



```

CPI = c(
  1068,1082,1106,1122,
  1142,1161,1186,1203,
  1218,1231,1253,1259,
  1267,1272,1280,1287,
  1299,1306,1319,1327
),

Inflation_YoY = c(
  NA,NA,NA,NA,
  6.928839,7.301294,7.233273,7.219251,
  6.654991,6.029285,5.649241,4.655029,
  4.022989,3.330626,2.154828,2.223987,
  2.525651,2.672956,3.046875,3.108003
),

Dwellings = c(
  1573,1985,2149,2007,
  2416,2056,2239,2187,
  1689,1723,1610,1937,
  1305,1766,1694,1779,
  1412,1614,1994,2296
),

AvgFloorArea = c(
  161,156,156,152,
  148,152,149,144,
  147,147,135,133,
  149,141,134,140,
  140,144,143,138
),

ValuePerSQM = c(
  2187,2145,2191,2276,
  2342,2445,2487,2617,
  2826,2836,2953,2991,
  2867,2871,2924,2934,
  2819,2860,2839,2966
),

TotalValue = c(
  554957428,663572561,736893624,694639931,
  836466019,764735580,828432622,825386776,
  701692504,717676865,640524498,771477576,
  556731415,712799164,664384547,729254267,
  556571853,666197380,810914068,936626292
),

TotalFloorArea = c(
  253757,309348,336280,305217,
  357167,312811,333061,315438,
  248305,253082,216870,257941,
  194181,248247,227211,248555,

```

```

    197423,232939,285633,315736
  ),

  ValuePerConsent = c(
    352802,334293,342901,346109,
    346219,371953,370001,377406,
    415448,416527,397841,398285,
    426614,403624,392199,409924,
    394173,412762,406677,407938
  )
)

master_q <- master_q %>%
  mutate(
    Year = as.numeric(substr(Quarter, 1, 4))
  ) %>%
  left_join(
    affordability_5yr,
    by = "Year"
  ) %>%
  arrange(Year, Quarter) %>%
  as.data.frame()

tbl(data.frame(Rows = nrow(master_q), Columns = ncol(master_q)))

```

Rows	Columns
20	14

```
tbl(data.frame(Column = names(master_q)))
```

Column
Quarter
HPI
NetMigration_NZ
CPI
Inflation_YoY
Dwellings
AvgFloorArea
ValuePerSQM
TotalValue
TotalFloorArea
ValuePerConsent
Year
ValueIncome
MortgageBurden

tbl(master_q)

Quarter	HPI	NetMigration_NZ	CPI	Inflation_YoY	Dwellings
2021Q1	3379	-1782	1068	NA	1573
2021Q2	3495	-6585	1082	NA	1985
2021Q3	3704	-10821	1106	NA	2149
2021Q4	3920	-14950	1122	NA	2007
2022Q1	3846	-19458	1142	6.929	2416
2022Q2	3681	-17692	1161	7.301	2056
2022Q3	3548	-5169	1186	7.233	2239
2022Q4	3484	24900	1203	7.219	2187
2023Q1	3382	74466	1218	6.655	1689
2023Q2	3349	108397	1231	6.029	1723
2023Q3	3404	132723	1253	5.649	1610
2023Q4	3464	127951	1259	4.655	1937
2024Q1	3470	100024	1267	4.023	1305
2024Q2	3398	70216	1272	3.331	1766
2024Q3	3381	42233	1280	2.155	1694
2024Q4	3414	23880	1287	2.224	1779
2025Q1	3412	14048	1299	2.526	1412
2025Q2	3375	11319	1306	2.673	1614
2025Q3	3376	12211	1319	3.047	1994
2025Q4	3392	18462	1327	3.108	2296

Quarter	AvgFloorArea	ValuePerSQM	TotalValue	TotalFloorArea
2021Q1	161	2187	554957428	253757
2021Q2	156	2145	663572561	309348
2021Q3	156	2191	736893624	336280
2021Q4	152	2276	694639931	305217
2022Q1	148	2342	836466019	357167
2022Q2	152	2445	764735580	312811
2022Q3	149	2487	828432622	333061
2022Q4	144	2617	825386776	315438
2023Q1	147	2826	701692504	248305
2023Q2	147	2836	717676865	253082
2023Q3	135	2953	640524498	216870
2023Q4	133	2991	771477576	257941
2024Q1	149	2867	556731415	194181
2024Q2	141	2871	712799164	248247
2024Q3	134	2924	664384547	227211
2024Q4	140	2934	729254267	248555
2025Q1	140	2819	556571853	197423
2025Q2	144	2860	666197380	232939
2025Q3	143	2839	810914068	285633
2025Q4	138	2966	936626292	315736

Quarter	ValuePerConsent	Year	ValueIncome	MortgageBurden
2021Q1	352802	2021	5.5	26.2
2021Q2	334293	2021	5.5	26.2
2021Q3	342901	2021	5.5	26.2
2021Q4	346109	2021	5.5	26.2
2022Q1	346219	2022	6.5	36.8
2022Q2	371953	2022	6.5	36.8
2022Q3	370001	2022	6.5	36.8
2022Q4	377406	2022	6.5	36.8
2023Q1	415448	2023	5.8	39.6
2023Q2	416527	2023	5.8	39.6
2023Q3	397841	2023	5.8	39.6
2023Q4	398285	2023	5.8	39.6
2024Q1	426614	2024	5.7	40.2
2024Q2	403624	2024	5.7	40.2
2024Q3	392199	2024	5.7	40.2
2024Q4	409924	2024	5.7	40.2
2025Q1	394173	2025	5.6	34.0
2025Q2	412762	2025	5.6	34.0
2025Q3	406677	2025	5.6	34.0
2025Q4	407938	2025	5.6	34.0

```
tbl(data.frame(
  Column = names(master_q),
  Missing_values = colSums(is.na(master_q)),
  row.names = NULL
))
```

Column	Missing_values
Quarter	0
HPI	0
NetMigration_NZ	0
CPI	0
Inflation_YoY	4
Dwellings	0
AvgFloorArea	0
ValuePerSQM	0
TotalValue	0
TotalFloorArea	0
ValuePerConsent	0
Year	0
ValueIncome	0
MortgageBurden	0

```
tbl(data.frame(
  Duplicate_quarter_position = anyDuplicated(master_q$Quarter)
))
```

Duplicate_quarter_position
0

```
tbl(data.frame(
  Year = unique(master_q$Year),
  Quarters_covered = as.vector(tapply(master_q$Quarter, master_q$Year, paste, collapse = ", ")),
  row.names = NULL
))
```

Year	Quarters_covered
2021	2021Q1, 2021Q2, 2021Q3, 2021Q4
2022	2022Q1, 2022Q2, 2022Q3, 2022Q4
2023	2023Q1, 2023Q2, 2023Q3, 2023Q4
2024	2024Q1, 2024Q2, 2024Q3, 2024Q4
2025	2025Q1, 2025Q2, 2025Q3, 2025Q4

```
write.csv(
  master_q,
  "master_housing_2021_2025.csv",
  row.names = FALSE
)
```

24.1 Data-quality finding

My completed master dataset contains **20 observations and 14 variables**, with one observation for every quarter from 2021Q1 through 2025Q4.

I found no duplicate quarters.

The four missing Inflation_YoY observations correspond to 2021 in this version of my master table.

25 24. Descriptive Statistics for the Quarterly Dataset

```
analysis_vars <- master_q %>%
  select(
    HPI,
    NetMigration_NZ,
    Inflation_YoY,
    Dwellings,
    AvgFloorArea,
    ValuePerSQM,
    ValuePerConsent
  )

descriptive_stats <- data.frame(
```

```

Variable = names(analysis_vars),
Mean = sapply(analysis_vars, mean, na.rm = TRUE),
Median = sapply(analysis_vars, median, na.rm = TRUE),
SD = sapply(analysis_vars, sd, na.rm = TRUE),
Min = sapply(analysis_vars, min, na.rm = TRUE),
Max = sapply(analysis_vars, max, na.rm = TRUE)
)

tbl(descriptive_stats)

```

Variable	Mean	Median	SD	Min
HPI	3493.700	3413.000	165.025	3349.000
NetMigration_NZ	34218.650	16255.000	50101.738	-19458.000
Inflation_YoY	4.672	4.339	2.000	2.155
Dwellings	1871.550	1858.000	302.208	1305.000
AvgFloorArea	145.450	145.500	7.702	133.000
ValuePerSQM	2668.800	2831.000	300.087	2145.000
ValuePerConsent	386184.800	396007.000	28733.295	334293.000

Variable	Max
HPI	3920.000
NetMigration_NZ	132723.000
Inflation_YoY	7.301
Dwellings	2416.000
AvgFloorArea	161.000
ValuePerSQM	2991.000
ValuePerConsent	426614.000

25.1 Interpretation

Across the 20 quarters, I found that:

- average HPI was approximately **3,493.7**;
- average net migration was approximately **34,219**;
- average year-on-year inflation for the available observations was approximately **4.67%**;
- average quarterly dwelling consents were approximately **1,872**;
- average floor area was approximately **145 square metres**;
- average value per square metre was approximately **\$2,669**;
- average value per consent was approximately **\$386,185**.

These statistics give me context but do not establish relationships between the variables.

26 25. Correlation Analysis Using Levels

```
correlation_matrix <- cor(
  analysis_vars,
  use = "pairwise.complete.obs",
  method = "pearson"
)

tbl(round(correlation_matrix, 3), row.names = TRUE)
```

Variable	HPI	NetMigration_NZ	Inflation_YoY	Dwellings
HPI	1.000	-0.512	0.576	0.547
NetMigration_NZ	-0.512	1.000	-0.029	-0.497
Inflation_YoY	0.576	-0.029	1.000	0.465
Dwellings	0.547	-0.497	0.465	1.000
AvgFloorArea	0.423	-0.553	0.573	0.127
ValuePerSQM	-0.654	0.690	-0.707	-0.398
ValuePerConsent	-0.695	0.661	-0.542	-0.517

Variable	AvgFloorArea	ValuePerSQM	ValuePerConsent
HPI	0.423	-0.654	-0.695
NetMigration_NZ	-0.553	0.690	0.661
Inflation_YoY	0.573	-0.707	-0.542
Dwellings	0.127	-0.398	-0.517
AvgFloorArea	1.000	-0.872	-0.626
ValuePerSQM	-0.872	1.000	0.927
ValuePerConsent	-0.626	0.927	1.000

My original level correlations included approximately:

- HPI and net migration: **-0.512**
- HPI and inflation: **0.576**
- HPI and dwelling consents: **0.547**
- HPI and value per square metre: **-0.654**
- value per square metre and value per consent: **0.927**

I interpret these relationships carefully because several variables change systematically through time.

I recognise that a correlation between two trending time series does not by itself demonstrate a causal economic relationship.

26.1 Individual correlation tests

```
level_cor_tests <- bind_rows(
  cor_row(
    cor.test(master_q$HPI, master_q$NetMigration_NZ),
    "HPI vs NetMigration_NZ"
  ),

```

```

cor_row(
  cor.test(master_q$HPI, master_q$Inflation_YoY),
  "HPI vs Inflation_YoY"
),
cor_row(
  cor.test(master_q$HPI, master_q$Dwellings),
  "HPI vs Dwellings"
),
cor_row(
  cor.test(master_q$HPI, master_q$ValuePerSQM),
  "HPI vs ValuePerSQM"
)
)

tbl(level_cor_tests, digits = 4)

```

Test	r	t	df	p_value	CI_Lower
HPI vs NetMigration_NZ	-0.5119	-2.5280	18	0.0210	-0.7781
HPI vs Inflation_YoY	0.5765	2.6396	14	0.0194	0.1131
HPI vs Dwellings	0.5473	2.7743	18	0.0125	0.1382
HPI vs ValuePerSQM	-0.6544	-3.6719	18	0.0017	-0.8506

Test	CI_Upper
HPI vs NetMigration_NZ	-0.0897
HPI vs Inflation_YoY	0.8339
HPI vs Dwellings	0.7968
HPI vs ValuePerSQM	-0.2983

My original results were approximately:

- HPI vs migration: $r = -0.512$, $p = 0.021$
- HPI vs inflation: $r = 0.576$, $p = 0.019$
- HPI vs dwellings: $r = 0.547$, $p = 0.013$
- HPI vs value per square metre: $r = -0.654$, $p = 0.0017$

Although these are statistically significant sample correlations, I note that the level series contain strong temporal patterns.

For this reason, my later analysis focuses more strongly on **quarterly HPI growth, changes and lagged variables**.

27 26. HPI Trend

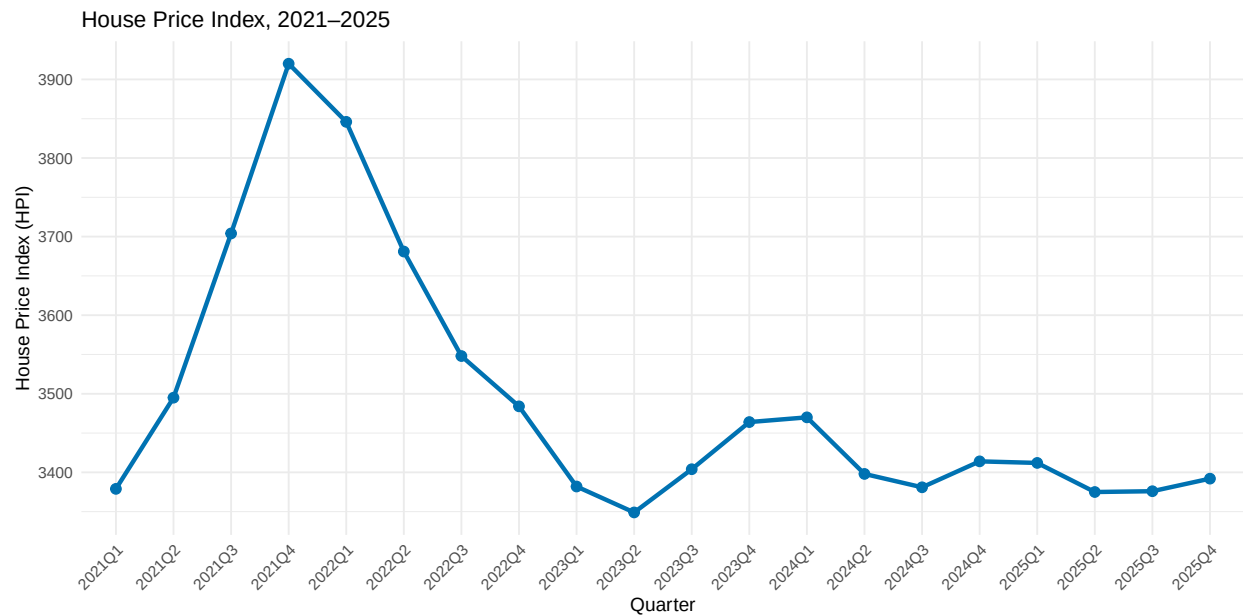
```

ggplot(
  master_q,
  aes(x = Quarter, y = HPI, group = 1)

```



```
) +
  geom_line(linewidth = 1.2, colour = "#0072B2") +
  geom_point(size = 2.5, colour = "#0072B2") +
  labs(
    title = "House Price Index, 2021–2025",
    x = "Quarter",
    y = "House Price Index (HPI)"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
)
```



27.1 Interpretation

I see HPI increase rapidly during 2021, reaching **3,920 in 2021Q4**.

I then see it decline through 2022 and reach **3,349 in 2023Q2**.

From 2023 onward, I see the index become considerably more stable, generally fluctuating in the mid-3300 to mid-3400 range.

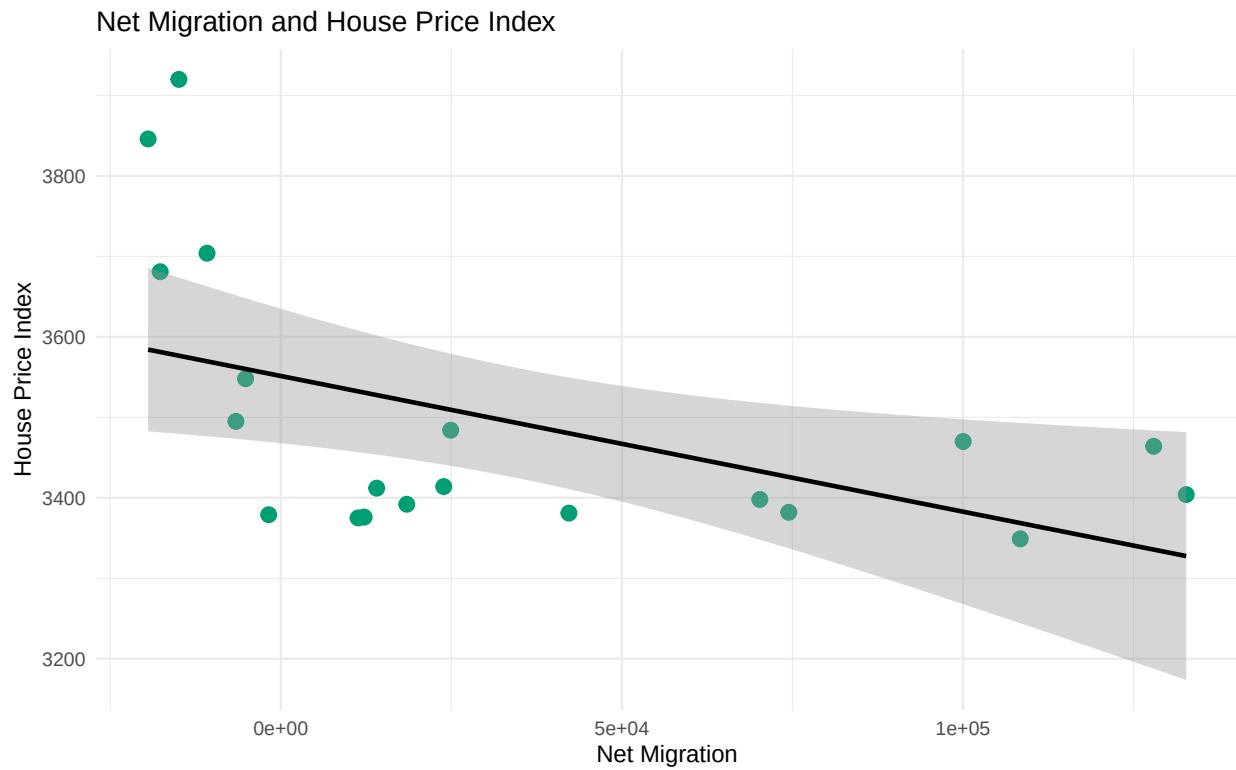
This suggests to me three broad phases: rapid appreciation, correction and relative stabilisation.

28 27. Scatterplots of HPI and Candidate Predictors

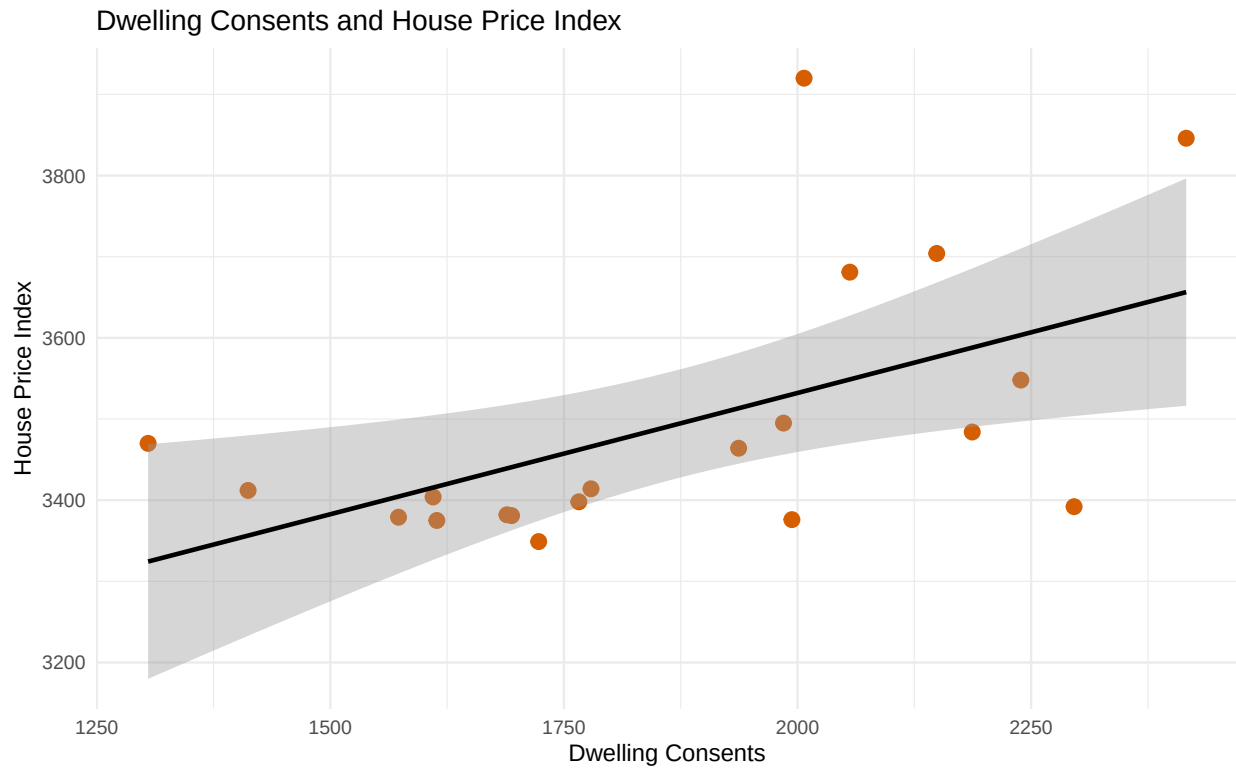
```
ggplot(master_q, aes(x = NetMigration_NZ, y = HPI)) +
  geom_point(size = 3, colour = "#009E73") +
  geom_smooth(method = "lm", se = TRUE, colour = "black") +
  labs(
    title = "Net Migration and House Price Index",

```

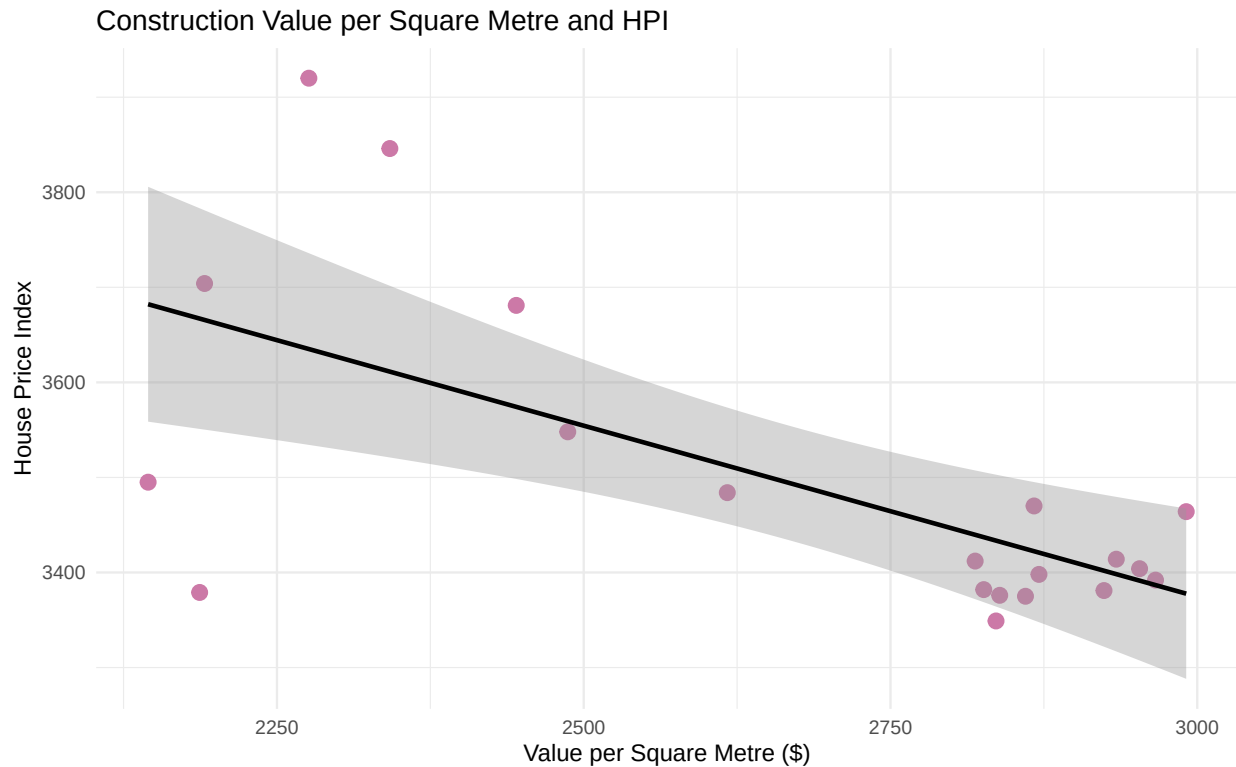
```
x = "Net Migration",
y = "House Price Index"
) +
theme_minimal()
```



```
ggplot(master_q, aes(x = Dwellings, y = HPI)) +
  geom_point(size = 3, colour = "#D55E00") +
  geom_smooth(method = "lm", se = TRUE, colour = "black") +
  labs(
    title = "Dwelling Consents and House Price Index",
    x = "Dwelling Consents",
    y = "House Price Index"
  ) +
  theme_minimal()
```



```
ggplot(master_q, aes(x = ValuePerSQM, y = HPI)) +
  geom_point(size = 3, colour = "#CC79A7") +
  geom_smooth(method = "lm", se = TRUE, colour = "black") +
  labs(
    title = "Construction Value per Square Metre and HPI",
    x = "Value per Square Metre ($)",
    y = "House Price Index"
  ) +
  theme_minimal()
```



These graphs visualise the level-data associations, but I do not interpret them as evidence that the explanatory variable caused the HPI movement.

29 28. Creating Quarterly Changes and Lagged Variables

To reduce reliance on relationships between trending levels, in the next stage I examine quarterly HPI growth and changes in the explanatory variables.

```
model_q <- master_q %>%
  arrange(Year, Quarter) %>%
  mutate(
    HPI_growth =
      100 * (HPI / lag(HPI) - 1),

    Migration_change =
      NetMigration_NZ - lag(NetMigration_NZ),

    Inflation_change =
      Inflation_YoY - lag(Inflation_YoY),

    Dwellings_change =
      100 * (Dwellings / lag(Dwellings) - 1),

    Cost_growth =
      100 * (ValuePerSQM / lag(ValuePerSQM) - 1),

    Migration_lag1 =
```

```

    lag(NetMigration_NZ, 1),

    Inflation_lag1 =
      lag(Inflation_YoY, 1),

    Dwellings_lag1 =
      lag(Dwellings, 1),

    Cost_lag1 =
      lag(ValuePerSQM, 1),

    HPI_growth_lag1 =
      lag(HPI_growth, 1)
  )
model_q %>%
  select(
    Quarter,
    HPI,
    HPI_growth,
    NetMigration_NZ,
    Inflation_YoY,
    Dwellings,
    ValuePerSQM
  ) %>%
  tbl()

```

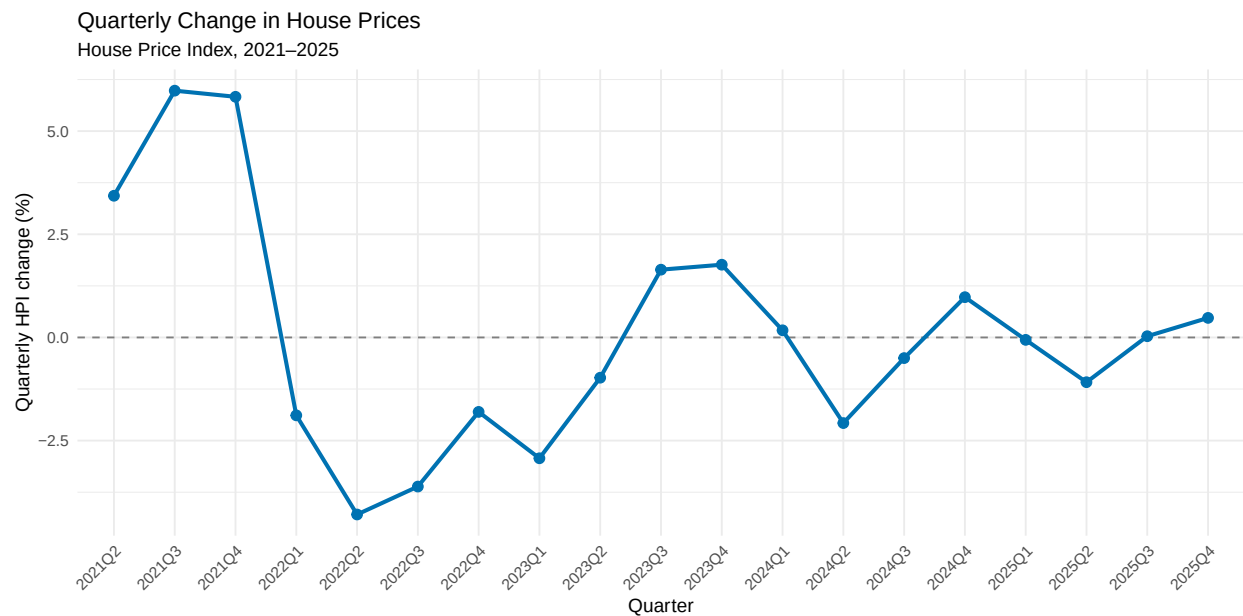
Quarter	HPI	HPI_growth	NetMigration_NZ	Inflation_YoY	Dwellings
2021Q1	3379	NA	-1782	NA	1573
2021Q2	3495	3.433	-6585	NA	1985
2021Q3	3704	5.980	-10821	NA	2149
2021Q4	3920	5.832	-14950	NA	2007
2022Q1	3846	-1.888	-19458	6.929	2416
2022Q2	3681	-4.290	-17692	7.301	2056
2022Q3	3548	-3.613	-5169	7.233	2239
2022Q4	3484	-1.804	24900	7.219	2187
2023Q1	3382	-2.928	74466	6.655	1689
2023Q2	3349	-0.976	108397	6.029	1723
2023Q3	3404	1.642	132723	5.649	1610
2023Q4	3464	1.763	127951	4.655	1937
2024Q1	3470	0.173	100024	4.023	1305
2024Q2	3398	-2.075	70216	3.331	1766
2024Q3	3381	-0.500	42233	2.155	1694
2024Q4	3414	0.976	23880	2.224	1779
2025Q1	3412	-0.059	14048	2.526	1412
2025Q2	3375	-1.084	11319	2.673	1614
2025Q3	3376	0.030	12211	3.047	1994
2025Q4	3392	0.474	18462	3.108	2296

Quarter	ValuePerSQM
2021Q1	2187
2021Q2	2145
2021Q3	2191
2021Q4	2276
2022Q1	2342
2022Q2	2445
2022Q3	2487
2022Q4	2617
2023Q1	2826
2023Q2	2836
2023Q3	2953
2023Q4	2991
2024Q1	2867
2024Q2	2871
2024Q3	2924
2024Q4	2934
2025Q1	2819
2025Q2	2860
2025Q3	2839
2025Q4	2966

30 29. Quarterly HPI Growth

```
ggplot(
  model_q %>% filter(!is.na(HPI_growth)),
  aes(x = Quarter, y = HPI_growth, group = 1)
) +
  geom_hline(
    yintercept = 0,
    linetype = "dashed",
    colour = "grey50"
  ) +
  geom_line(
    linewidth = 1.1,
    colour = "#0072B2"
  ) +
  geom_point(
    size = 2.5,
    colour = "#0072B2"
  ) +
  labs(
    title = "Quarterly Change in House Prices",
    subtitle = "House Price Index, 2021-2025",
    x = "Quarter",
    y = "Quarterly HPI change (%)"
  ) +
  theme_minimal() +
  theme(
```

```
axis.text.x = element_text(angle = 45, hjust = 1)
)
```



30.1 Interpretation

I find that quarterly growth shows the turning points more clearly than the HPI level.

I see strong positive growth during 2021 followed by several quarters of negative growth during 2022 and early 2023.

I then see growth become much closer to zero during 2024–2025.

For this reason, I use `HPI_growth` as the main dependent variable in my regression analysis.

31 30. Correlations Using Quarterly Changes

```
change_vars <- model_q %>%
  select(
    HPI_growth,
    Migration_change,
    Inflation_change,
    Dwellings_change,
    Cost_growth
  )

change_cor <- cor(
  change_vars,
  use = "pairwise.complete.obs",
  method = "pearson"
)
```

```
tbl(round(change_cor, 3), row.names = TRUE)
```

Variable	HPI_growth	Migration_change	Inflation_change
HPI_growth	1.000	-0.233	-0.211
Migration_change	-0.233	1.000	0.133
Inflation_change	-0.211	0.133	1.000
Dwellings_change	0.114	-0.204	0.000
Cost_growth	-0.183	0.625	-0.020

Variable	Dwellings_change	Cost_growth
HPI_growth	0.114	-0.183
Migration_change	-0.204	0.625
Inflation_change	0.000	-0.020
Dwellings_change	1.000	-0.071
Cost_growth	-0.071	1.000

31.1 Interpretation

I found that the correlations using changes were much weaker than many of the correlations using levels. My original results included:

- HPI growth and migration change: **-0.233**
- HPI growth and inflation change: **-0.211**
- HPI growth and dwelling change: **0.114**
- HPI growth and construction-value growth: **-0.183**

This is important because it suggests to me that some of the stronger level correlations were at least partly related to the way the variables moved through time rather than strong quarter-to-quarter relationships.

One notable correlation I found among the explanatory changes was approximately **0.625 between migration change and construction-value growth**.

32 31. Simple Regression Models

32.1 31.1 Migration Model

32.1.1 Reason for model

I include migration because it may affect housing demand.

With this model I test whether the previous quarter's net-migration level is associated with current HPI growth.


```
model_migration <- lm(
  HPI_growth ~ Migration_lag1,
  data = model_q
)

coef_tbl(model_migration)
```

Term	Estimate	Std_Err	t_value	p_value	CI_Low
(Intercept)	-0.02776	0.8092	-0.03431	0.973	-1.735
Migration_lag1	2.422e-06	1.326e-05	0.1827	0.8572	-2.555e-05

Term	CI_High
(Intercept)	1.68
Migration_lag1	3.04e-05

```
fit_tbl(model_migration)
```

N	R2	Adj_R2	Resid_SE	F_stat	p_value
19	0.00196	-0.05675	2.888	0.03338	0.8572

32.2 Interpretation

My simple migration regression did not provide strong evidence that lagged migration alone explained quarterly HPI growth in this short sample.

I do not interpret the result as evidence that migration has no effect on housing markets.

My model contains only a small number of quarters and does not control for the many other economic factors that can influence house prices.

32.3 31.2 Housing Supply Model

32.3.1 Reason for model

I use dwelling consents as an indicator of future housing supply.

```
model_supply <- lm(
  HPI_growth ~ Dwellings_lag1,
  data = model_q
)

coef_tbl(model_supply)
```

Term	Estimate	Std_Err	t_value	p_value	CI_Low
(Intercept)	2.994	4.29	0.6979	0.4947	-6.057
Dwellings_lag1	-0.001588	0.002293	-0.6927	0.4979	-0.006426

Term	CI_High
(Intercept)	12.05
Dwellings_lag1	0.003249

```
fit_tbl(model_supply)
```

N	R2	Adj_R2	Resid_SE	F_stat	p_value
19	0.02745	-0.02976	2.851	0.4798	0.4979

32.4 Interpretation

I found that lagged dwelling consents were not a strong individual predictor of quarterly HPI growth in the regression specification I used.

I do not take this to mean that housing supply is unimportant.

I note that consents are approvals rather than completed dwellings, and the effect of new housing supply may take longer than one quarter to appear.

32.5 31.3 Construction-Value Growth Model

```
model_cost <- lm(
  HPI_growth ~ Cost_growth,
  data = model_q
)

coef_tbl(model_cost)
```

Term	Estimate	Std_Err	t_value	p_value	CI_Low	CI_High
(Intercept)	0.3346	0.7455	0.4489	0.6592	-1.238	1.908
Cost_growth	-0.1671	0.2178	-0.7673	0.4534	-0.6265	0.2923

```
fit_tbl(model_cost)
```

N	R2	Adj_R2	Resid_SE	F_stat	p_value
19	0.03347	-0.02338	2.842	0.5888	0.4534

32.6 Interpretation

I also found that quarterly growth in the value of consented building work per square metre was not a strong individual predictor of quarterly HPI growth in this sample.

Again, I do not interpret ValuePerSQM as a complete construction-cost index.

33 32. Small Combined Regression

In the next model I jointly examine migration, dwelling consents and construction-value growth.

```
model_combined <- lm(
  HPI_growth ~
    Migration_lag1 +
    Dwellings_lag1 +
    Cost_growth,
  data = model_q
)

coef_tbl(model_combined)
```

Term	Estimate	Std_Err	t_value	p_value	CI_Low
(Intercept)	2.519	5.999	0.42	0.6805	-10.27
Migration_lag1	-3.243e-06	1.602e-05	-0.2025	0.8423	-3.738e-05
Dwellings_lag1	-0.001159	0.003191	-0.3633	0.7214	-0.00796
Cost_growth	-0.1233	0.2772	-0.4449	0.6628	-0.7141

Term	CI_High
(Intercept)	15.31
Migration_lag1	3.089e-05
Dwellings_lag1	0.005642
Cost_growth	0.4675

```
fit_tbl(model_combined)
```

N	R2	Adj_R2	Resid_SE	F_stat	p_value
19	0.04211	-0.1495	3.012	0.2198	0.8811

33.1 Interpretation

My original combined model produced approximately:

$$R^2 = 0.042$$

with:

$$\text{Adjusted } R^2 = -0.150$$

and an overall:

$$p = 0.881$$

I found that none of the three predictors was statistically significant.

My model therefore explained very little of the observed quarter-to-quarter HPI growth.

This was an important result for me because it suggested that another feature of the HPI series itself might be more informative.

34 33. HPI Persistence Regression

34.1 Why this model is important

I considered that housing-market growth may display short-run persistence.

In other words, strong growth in one quarter may be followed by stronger growth in the next quarter, while weak growth may also persist.

I test this directly with the following simple model.

```
model_persistence <- lm(  
  HPI_growth ~ HPI_growth_lag1,  
  data = model_q  
)  
  
coef_tbl(model_persistence)
```

Term	Estimate	Std_Err	t_value	p_value	CI_Low	CI_High
(Intercept)	-0.1507	0.526	-0.2864	0.7782	-1.266	0.9644
HPI_growth_lag1	0.5957	0.1874	3.18	0.005822	0.1985	0.9929

```
fit_tbl(model_persistence)
```

N	R2	Adj_R2	Resid_SE	F_stat	p_value
18	0.3872	0.3489	2.231	10.11	0.005822

34.2 Main Results

My original fitted model produced approximately:

$$\beta_{HPI\ growth\ lag1} = 0.596$$

$$p = 0.0058$$

$$R^2 = 0.387$$

$$Adjusted\ R^2 = 0.349$$

34.3 Interpretation

I found that lagged HPI growth was statistically significant at the 1% level.

I interpret the coefficient of approximately **0.596** as meaning that a one-percentage-point higher HPI growth rate in the previous quarter was associated with approximately **0.60 percentage points higher HPI growth in the current quarter**, on average within this sample.

My model explains approximately **38.7% of the observed variation** in quarterly HPI growth.

This is the clearest regression result I found.

Main regression finding: I found evidence of short-run persistence in quarterly house-price growth during 2021–2025. Previous-quarter HPI growth was positively and statistically significantly associated with current-quarter HPI growth.

This is an association, and I do not interpret it as a causal effect.

35 34. Mixed Regression — Persistence and Migration

With this model I test whether migration adds explanatory information after controlling for existing house-price momentum.

```
model_mixed <- lm(
  HPI_growth ~
    HPI_growth_lag1 +
    Migration_lag1,
  data = model_q
)

coef_tbl(model_mixed)
```

Term	Estimate	Std_Err	t_value	p_value	CI_Low
(Intercept)	-0.4125	0.6646	-0.6207	0.5441	-1.829
HPI_growth_lag1	0.6022	0.191	3.153	0.00656	0.1952

Term	Estimate	Std_Err	t_value	p_value	CI_Low
Migration_lag1	7.053e-06	1.06e-05	0.6651	0.5161	-1.555e-05

Term	CI_High
(Intercept)	1.004
HPI_growth_lag1	1.009
Migration_lag1	2.965e-05

```
fit_tbl(model_mixed)
```

N	R2	Adj_R2	Resid_SE	F_stat	p_value
18	0.4048	0.3254	2.271	5.1	0.02043

35.1 Interpretation

My main purpose with this model is to compare the migration coefficient with the simple migration regression.

If migration becomes weak once previous HPI growth is controlled for, I take that to suggest that short-run price momentum is more informative within this particular sample.

I still treat the result cautiously because my dataset is small.

36 35. Demand and Supply Regression

This model lets me jointly consider a demand-side indicator and a future supply indicator.

```
model_demand_supply <- lm(
  HPI_growth ~
    Migration_lag1 +
    Dwellings_lag1,
  data = model_q
)

coef_tbl(model_demand_supply)
```

Term	Estimate	Std_Err	t_value	p_value	CI_Low
(Intercept)	3.557	5.386	0.6605	0.5183	-7.861
Migration_lag1	-2.849e-06	1.559e-05	-0.1828	0.8573	-3.589e-05
Dwellings_lag1	-0.001839	0.00273	-0.6735	0.5102	-0.007627

Term	CI_High
(Intercept)	14.98
Migration_lag1	3.019e-05
Dwellings_lag1	0.003949

```
fit_tbl(model_demand_supply)
```

N	R2	Adj_R2	Resid_SE	F_stat	p_value
19	0.02948	-0.09184	2.935	0.243	0.7871

36.1 Interpretation

With this model I examine whether lagged migration and dwelling consents jointly explain HPI growth.

I do not interpret either variable causally because both may themselves respond to broader economic conditions.

37 36. Full Multivariable Regression

37.1 Reason for model

With the full model I test whether migration, dwelling consents and construction-value growth provide additional explanatory information after accounting for previous HPI growth.

```
model_full <- lm(
  HPI_growth ~
    HPI_growth_lag1 +
    Migration_lag1 +
    Dwellings_lag1 +
    Cost_growth,
  data = model_q
)

coef_tbl(model_full)
```

Term	Estimate	Std_Err	t_value	p_value	CI_Low
(Intercept)	0.6694	5.103	0.1312	0.8976	-10.35
HPI_growth_lag1	0.6079	0.2075	2.93	0.01172	0.1596
Migration_lag1	6.224e-06	1.392e-05	0.447	0.6622	-2.386e-05
Dwellings_lag1	-0.0006128	0.002658	-0.2306	0.8212	-0.006354
Cost_growth	0.04907	0.2329	0.2106	0.8364	-0.4542

Term	CI_High
(Intercept)	11.69
HPI_growth_lag1	1.056
Migration_lag1	3.631e-05
Dwellings_lag1	0.005128
Cost_growth	0.5523

```
fit_tbl(model_full)
```

N	R2	Adj_R2	Resid_SE	F_stat	p_value
18	0.4079	0.2258	2.433	2.239	0.121

37.2 Main Results

My original full model produced approximately:

- Lagged HPI growth coefficient: **0.608**
- Lagged HPI growth p-value: **0.0117**
- Migration p-value: **0.662**
- Dwelling consents p-value: **0.821**
- Construction-value growth p-value: **0.836**
- $R^2 \approx 0.408$
- Adjusted $R^2 \approx 0.226$
- Overall model p-value: **0.121**

37.3 Interpretation

I found that lagged HPI growth remained statistically significant after I added migration, dwelling consents and construction-value growth to the model.

The coefficient remained close to that in my simple persistence model, at approximately **0.608**.

By contrast, I found that migration, dwelling consents and construction-value growth were not individually statistically significant.

I see the full model's R^2 increase only slightly from approximately **0.387 to 0.408**, while adjusted R^2 declined from approximately **0.349 in the persistence model to 0.226**.

This suggests to me that the additional variables provided relatively little improvement in explanatory performance once the penalty for additional model complexity is considered.

I do **not** take these findings to establish that migration, housing supply or construction conditions have no effect on the housing market.

I can say only that these variables did not provide strong additional explanatory information in this particular model, period and small sample.

38 37. Model Comparison

```
model_comparison <- AIC(
  model_persistence,
  model_combined,
  model_full
)

tbl(model_comparison, digits = 2, row.names = TRUE)
```

Variable	df	AIC
model_persistence	3	83.86
model_combined	5	101.32
model_full	6	89.24

My original AIC values were approximately:

Model	AIC
Persistence	83.86
Combined	101.32
Full	89.24

I found that the persistence model had the lowest AIC among these specifications.

Because AIC rewards fit while penalising unnecessary complexity, this gives me additional support for using the simpler persistence model as the most economical description among the models tested.

I do not interpret this as proving that persistence is the only process affecting house prices.

39 38. Model Sample Sizes

```
tbl(data.frame(
  Model = c("Persistence", "Combined", "Full"),
  Observations = c(
    nobs(model_persistence),
    nobs(model_combined),
    nobs(model_full)
  )
))
```

Model	Observations
Persistence	18
Combined	19
Full	18

My original models used approximately:

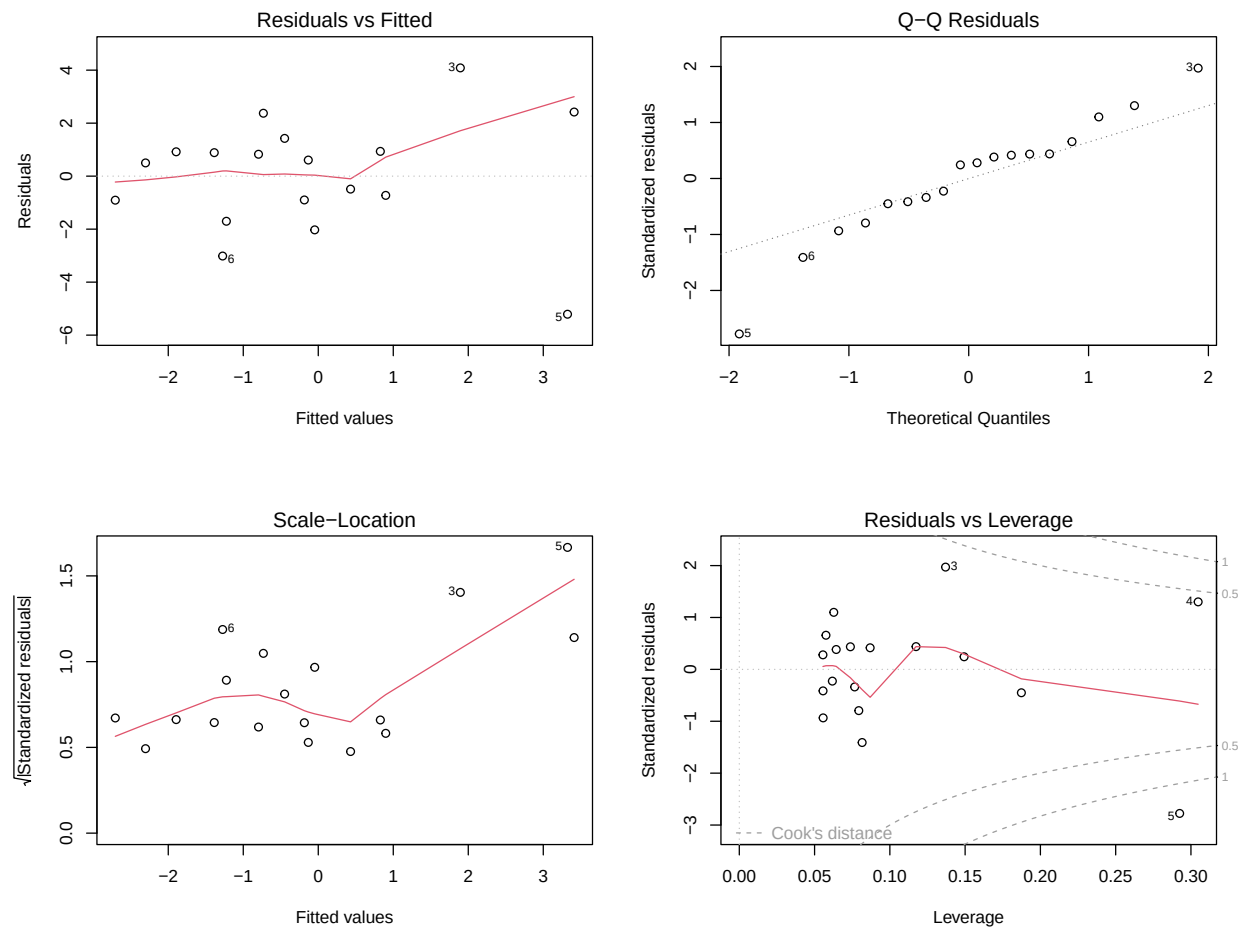
- Persistence model: **18 observations**
- Combined model: **19 observations**
- Full model: **18 observations**

I recognise this is a very small sample.

The small number of observations limits my statistical power and means the multivariable coefficients can be sensitive to individual quarters.

40 39. Regression Diagnostics

```
par(mfrow = c(2, 2))
plot(model_persistence)
```



```
par(mfrow = c(1, 1))
```

40.1 Interpretation

I inspect the diagnostic plots for:

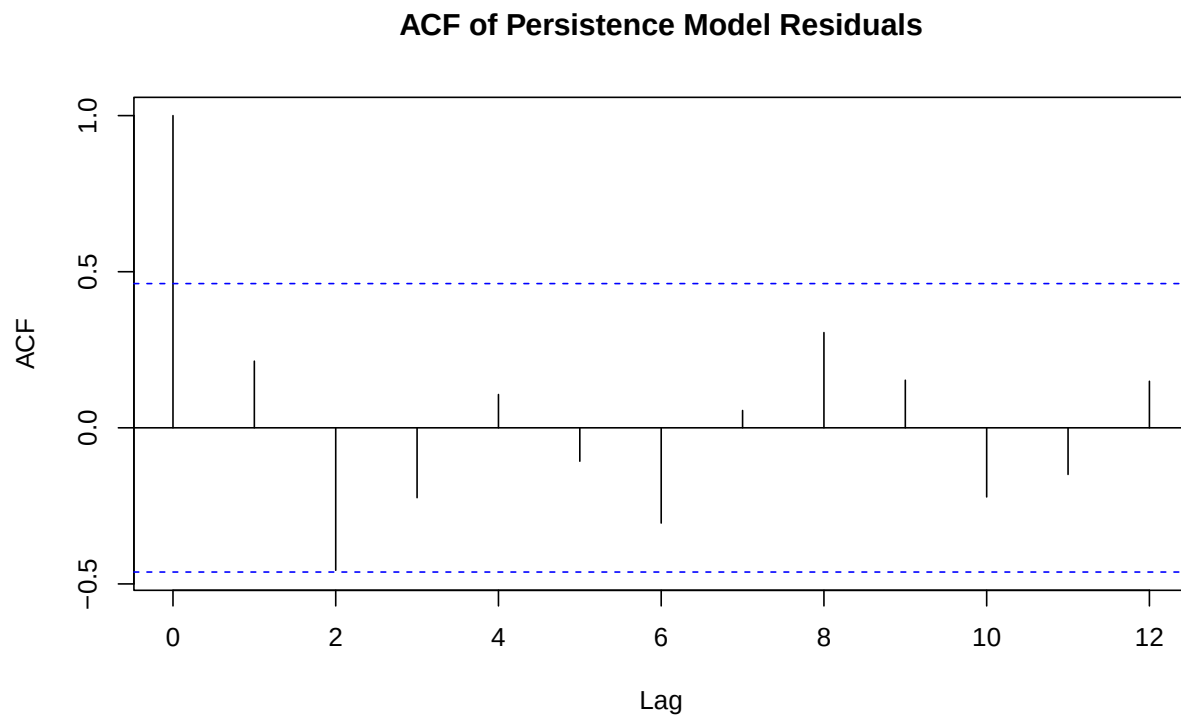
- non-linearity;
- unequal residual variance;
- unusually influential observations;
- major departures from normal residual behaviour.

My original inspection suggested several potentially influential observations and some indication that residual variance may be larger at higher fitted values.

Given the very small sample, I treat these diagnostics seriously.

41 40. Residual Autocorrelation

```
acf(
  residuals(model_persistence),
  main = "ACF of Persistence Model Residuals"
)
```



```
dw_test <- dwtest(
  model_persistence,
  alternative = "greater"
)

tbl(
  data.frame(
    DW_statistic = unname(dw_test$statistic),
    p_value = dw_test$p.value,
  )
)
```

```

    Alternative = dw_test$alternative
  ),
  digits = 4
)

```

DW_statistic	p_value	Alternative
1.3587	0.0559	true autocorrelation is greater than 0

41.1 Interpretation

My original Durbin-Watson test produced:

$$DW = 1.359$$

with:

$$p = 0.056$$

for the one-sided test of positive residual autocorrelation.

I therefore find positive residual autocorrelation **not statistically significant at the conventional 5% level**, although the p-value is close to the threshold.

I do not describe the result as proof that autocorrelation is absent.

It instead indicates that my available sample did not provide sufficiently strong evidence to reject the null at the 5% level.

42 41. Main Findings from the Regression Analysis

My regression analysis produces a fairly clear overall story.

First, I found several strong relationships when I analysed the variables in **levels**. For example, HPI was negatively correlated with net migration and value per square metre and positively correlated with inflation and dwelling consents.

However, I treat these level relationships cautiously because many of the variables moved systematically through time.

When I shifted to **quarterly changes**, the correlations became considerably weaker.

My simple regression containing migration, dwelling consents and construction-value growth explained very little quarterly HPI growth.

My strongest result appeared when I included previous HPI growth.

I found that lagged HPI growth had a coefficient of approximately **0.596** in the simple persistence model and remained around **0.608** after I included the other explanatory variables.

The coefficient was statistically significant in both specifications.

Therefore, within this 2021–2025 sample, I identify **short-run persistence in HPI growth as the clearest statistical relationship**.

43 42. How These Results Help the Project

My analysis helps me answer the broader research questions in several ways.

For the question concerning **factors associated with housing affordability**, my Canterbury affordability EDA shows that affordability changed particularly strongly around 2021–2024.

The divergence between the value-to-income multiple and mortgage burden also shows me that affordability is multidimensional. House values relative to income began improving after 2022, while mortgage burden continued increasing through 2024.

My dwelling-consent analysis provides evidence that Canterbury housing development activity also changed substantially during the same period, but I do not see a simple immediate relationship with affordability in the descriptive comparisons.

My quarterly regression adds another result: over 2021–2025, quarter-to-quarter HPI growth was more strongly associated with its own previous-quarter growth than with the migration, consenting and construction-value measures I tested.

This suggests to me that the recent housing market cannot be described adequately using one explanatory factor.

44 43. What the Analysis Does Not Show

I do **not** interpret my regressions as causal models.

In particular, my analysis does not establish that:

- migration causes or does not cause house-price growth;
- dwelling consents cause or do not cause house-price changes;
- construction conditions have no effect on housing;
- previous HPI growth itself causes future HPI growth.

My regression results identify associations conditional on the variables and time period included in the models.

Several predictors may also respond jointly to broader economic conditions.

45 44. Limitations

The most important limitation I face is the small sample.

My quarterly regression covers only **2021–2025**, giving me approximately 18–20 usable observations depending on the model.

This period was also economically unusual, containing rapid house-price growth, a substantial correction, high inflation and major changes in migration and financing conditions.

I therefore do not automatically generalise my results to other periods.

I also note that dwelling consents measure approvals rather than completed housing supply.

ValuePerSQM represents the value of consented building work, and I do not interpret it as a complete construction-cost index.

I also note that net migration and CPI are New Zealand-level measures rather than Canterbury-specific measures.

The annual Canterbury affordability variables are repeated within years when I join them to the quarterly master dataset. I therefore use them mainly for context rather than as independent quarterly outcomes.

My regression analysis identifies associations rather than causal effects.

46 45. Overall Conclusion

My analysis shows that Canterbury housing affordability changed substantially during the period surrounding 2021–2024.

I found that the house value-to-income multiple deteriorated sharply during 2021–2022 and subsequently improved, while mortgage burden continued increasing through 2024 before declining.

I found that Canterbury dwelling consenting activity peaked in 2022, declined during 2023–2024 and recovered partly during 2025.

I also found that the value of consented residential building work per square metre increased substantially between 2021 and 2023.

In my quarterly HPI regression analysis, relationships that appeared relatively strong in the level data became much weaker when I considered quarterly changes.

My clearest regression finding was **short-run persistence in HPI growth**.

I found that previous-quarter HPI growth was positively associated with current-quarter HPI growth, with a coefficient of approximately **0.596** and a p-value of approximately **0.0058** in the simple persistence model.

The relationship remained statistically significant after I added lagged migration, dwelling consents and construction-value growth.

I found that those additional predictors were not individually statistically significant in the full model and produced only a small increase in R^2 .

Given the small sample, I interpret these results cautiously.

Overall, the evidence suggests to me that recent housing-market movements were complex and that no single demand, supply or construction variable I examined provides a complete explanation of the changes observed during 2021–2025.